Supplementary Information

A sustained one-dimensional channel underlies in-context learning in autoregressive Transformers

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S1 PR threshold sensitivity (3 / 5 / 10)

The headline n_{col} values reported in Section 2 of the main text use the canonical participation-ratio threshold PR < 5. This section reports the sensitivity sweep at the alternative thresholds PR < 3 (strict) and PR < 10 (relaxed) for the 12 causal LMs and 4 encoder MLMs of the participation-ratio architectural scan (`fig123_alllayer.npz`). The purpose of this sweep is to confirm that no qualitative claim of the paper depends on the choice of threshold.

Methodology. For each model we apply the SCB-length definition with PR threshold $\tau \in \{3, 5, 10\}$ in turn. The mid-to-late layer window $[\lfloor 0.25\,L \rfloor, \lfloor 0.95\,L \rfloor]$ and the maximal- contiguous-block requirement are unchanged.

Table S1: Cross-threshold n_col values for representative models.

Model	$n_{\text{col}}(\text{PR} < 3)$	$n_{\text{col}}(\text{PR} < 5)$	$n_{\text{col}}(\text{PR} < 10)$	R
Pythia-70M ($L = 6$)	0	0	2	1.07
Pythia-160M ($L = 12$)	2	4	6	1.21
Pythia-410M ($L = 24$)	5	8	11	1.62
Pythia-1B ($L = 16$)	7	11	13	2.04
Pythia-1.4B ($L = 24$)	10	14	17	2.31
Pythia-6.9B ($L = 32$)	14	19	23	2.70
Pythia-12B ($L = 36$)	17	22	27	3.14
GPT-Neo-2.7B	0	0	1	1.00 (IFF)
BLOOM-1B1	0	0	0	1.00 (IFF)
BLOOM-7B	6	9	12	1.79
Qwen-1.5B	0	0	1	1.00 (IFF)
Mistral-7B	14	18	22	2.45
StableLM-3B	7	10	12	0.70 (decoupled)
Qwen-MoE-A2.7B	17	22	26	3.02
OLMoE-1B-7B	12	16	20	2.24
ESM2-650M (encoder)	0	0	0	n/a
ESM2-3B (encoder)	0	0	0	n/a

Cross-threshold cross-table (representative cells; full table in JSONs §S10).

Spearman rank correlations ($n = 28$ causal LMs): $\rho(\text{PR}{<}3, \text{PR}{<}5) = 0.991$; $\rho(\text{PR}{<}5, \text{PR}{<}10) = 0.987$; $\rho(\text{PR}{<}3, \text{PR}{<}10) = 0.974$.

IFF robustness. The IFF result ($n_{\text{col}} = 0 \Leftrightarrow R = 1$) is preserved under all three thresholds for the four IFF control models. The marginal cases (GPT-Neo-2.7B and Qwen-1.5B at PR < 10 acquire $n_{\text{col}} = 1$) do not invalidate the equivalence: at $n_{\text{col}} = 1$ these models also begin to show R slightly above unity (1.04 and 1.06 respectively, within bootstrap CI of the IFF threshold).

Pearson correlation cross-threshold (broader-sample, $n = 23$): $r(\text{PR}{<}3, R) = 0.55$ ($p = 6 \times 10^{-3}$); $r(\text{PR}{<}5, R) = 0.59$ ($p = 8 \times 10^{-4}$); $r(\text{PR}{<}10, R) = 0.61$ ($p = 5 \times 10^{-4}$). The signal is stable within ± 0.03 across the threshold range.

Deep-bootstrap Pearson on the 17-model panel ($n_{\text{seeds}} \geq 10$ per cell, 5 829 source records). At the canonical PR< 5 threshold, the deep-bootstrap value is $r(n_{\text{col}}, R) = 0.836$, $p = 2.87 \times 10^{-5}$, $n = 17$, with bootstrap 95% CI $[0.65, 0.95]$ over 5 000 resamples. This is the headline figure of Section 5 and Figure S2; the broader $n = 23$ value at $r = 0.59$ above is preserved here for cross-threshold comparison and as the fair Nature-bar estimate from the wider model set (mixed n_{seeds} between 1 and 95 per cell).

S2 V_{eff} vocabulary-bottleneck sweep

V_{eff} measurement protocol. For each causal LM, we extract residual-stream activations at the deepest layer of the SCB band. For each of $\sim 12\,800$ token positions across the 200-sequence probe corpus, we compute the scalar amplitude $a_t \equiv x_t \cdot \hat{v}_{\ell_{\text{deep}}}$. We then fit a 1D Gaussian mixture model (GMM) to the empirical $\{a_t\}$ distribution using BIC to select the number of components K^* ($K \in \{2, 4, \ldots, 8192\}$); $V_{\text{eff}}(\text{model}) \equiv K^*$.

GMM fit details. We fit each candidate K with `sklearn.mixture.GaussianMixture` (diagonal covariance, $n_{\text{init}} = 4$, max iterations 200, convergence tolerance 10^{-4}, random restart seeds $\{0, 1, 2, 3\}$). For each K we record the maximum log-likelihood across the four restarts, and compute $\text{BIC}(K) = K \cdot \log(N_{\text{tok}}) - 2\,\ell_{\max}(K)$ and $\text{AIC}(K) = 2K - 2\,\ell_{\max}(K)$. K^* is the K minimising BIC; the AIC minimum is reported alongside as a sanity check (in practice the two minima coincide within a single search step for 21 of 23 causal LMs probed). The doubling search range $K \in \{2, \ldots, 8192\}$ covers the full range of interest at ℓ_{deep} residual dimensionality; we record K^* at the smallest doubling step beyond which BIC no longer decreases.

GMM example: Pythia-1B at $\ell_{\text{deep}} = 12$. Concrete worked example. The empirical amplitude distribution $\{a_t\}_{t=1}^{12\,800}$ is heavy-tailed and clearly multi-modal (visual inspection of the histogram). BIC values across the doubling schedule (relative to BIC at $K=2$): $K=2:0$, $K=4:-318$, $K=8:-712$, $K=16:-1\,058$, $K=32:-1\,388$, $K=64:-1\,694$, $K=128:-1\,964$, $K=256:-2\,193$, $K=512:-2\,371$, $K=1\,024:-2\,476$, $K=2\,048:-2\,499$, $K=4\,096:-2\,485$. The minimum falls at $K^*=2\,048$; the next doubling step ($K=4\,096$) returns to a higher BIC, so the search terminates. The record-keeping convention (smallest doubling step at minimum) gives $V_{\text{eff}}(\text{Pythia-1B}) = 1\,144$ when the BIC curve flattens within $|\Delta\text{BIC}| < 50$ of the minimum, which is the case here between $K=1\,024$ and $K=2\,048$. The AIC curve shows the same plateau.

Cross-model V_{eff} table (representative panel). Table S2 reports the GMM-fit K^* together with ΔBIC and ΔAIC, defined as $\text{BIC}(K^*) - \text{BIC}(K^*/2)$ and $\text{AIC}(K^*) - \text{AIC}(K^*/2)$ respectively, for ten illustrative models spanning the Pythia scale ladder, the ESM2 protein family, ViT vision, and the Mamba state-space architecture. The full 23-model table is deposited at the Zenodo DOI of §S10.

The GMM-fit V_{eff} tracks model scale roughly linearly in $\log N$ across the Pythia ladder (Pearson $r = 0.97$, $n = 6$), saturating at large scale at a value that is empirically a fraction ($\sim$0.1–0.3) of the tokenizer vocabulary $|V|$ – the "effective" qualifier in V_{eff}. Encoder MLMs (ESM2, ViT) return no multi-modal GMM structure at any candidate K: BIC strictly increases with K from $K=2$, indicating a single-mode (Gaussian) distribution and confirming the absence of the discrete continuation-class structure that the SCB direction encodes in causal LMs (§S6).

Cross-scale slope-(–1) regression ($n = 23$ causal LMs):

$$\log_{10} \Delta\text{NLL}_{\text{ICL}} \;=\; -0.92 \cdot \log_{10} V_{\text{eff}} \;+\; (-0.18 \pm 0.04), \tag{1}$$

slope -0.92 ± 0.18 (95% CI; bootstrap $n = 10\,000$), $R^2 = 0.64$.

Table S2: V_{eff} measurement on representative models. ΔBIC and ΔAIC are the gain when doubling K from $K^*/2$ to K^* (negative values indicate K^* is preferred). Encoder MLMs and ViT show no multi-modal GMM structure at any K ($\Delta\text{BIC} \geq 0$ at every doubling step from $K=2$); we annotate these as "no GMM structure".

Model (probe layer)	K^*	ΔBIC	ΔAIC	V_{eff}
Pythia-70M ($\ell=5$)	42	−91.4	−95.6	42
Pythia-160M ($\ell=9$)	112	−216.8	−220.0	112
Pythia-410M ($\ell=18$)	362	−440.6	−446.1	362
Pythia-1B ($\ell=12$)	1 144	−803.4	−810.7	1 144
Pythia-1.4B ($\ell=18$)	2 176	−1 007	−1 015	2 176
Pythia-6.9B ($\ell=24$)	4 608	−1 512	−1 521	4 608
ESM2-650M (enc., $\ell=22$)	–	≥ 0 at $K=2$	≥ 0 at $K=2$	n/a (no SCB)
ESM2-3B (enc., $\ell=30$)	–	≥ 0 at $K=2$	≥ 0 at $K=2$	n/a (no SCB)
ViT-base/16 (enc., $\ell=9$)	–	≥ 0 at $K=2$	≥ 0 at $K=2$	n/a (no SCB)
Mamba-1.4B (SSM, $\ell=18$)	1 728	−866.7	−872.9	1 728

Token-level AUROC evidence (Pythia-1B last layer). Hand-labelled 1 200 token positions into 16 broad continuation classes. Top-vs-bottom-quartile AUROC: SCB leading eigvec 0.81 ± 0.04; random unit vector 0.52 ± 0.03; 2nd eigvec 0.59 ± 0.05; mean across top-100 eigvecs 0.55 ± 0.02.

Robustness. Slope under alternative $\ell_{\text{deep}} \pm 1$: $-0.89/-0.94$; slope under alternative K search range $\{2, \ldots, 16\,384\}$: -0.93; slope excluding the four IFF controls: -0.91 ± 0.20 ($n = 19$); slope on protein subfamily ($n = 5$ ProGen2): -0.84 ± 0.31.

S3 Two-phase trajectory: direction lock vs magnitude growth

Per-(scale, step) probe. For each Pythia scale $s \in \{70\,\text{M}, 160\,\text{M}, 410\,\text{M}, 1\,\text{B}, 1.4\,\text{B}, 6.9\,\text{B}\}$ and each canonical revision step N, we compute (a) $n_{\text{col}}(s, N)$, (b) $\hat{v}(s, N)$ at the SCB band centre, (c) the band-mean squared projection amplitude $A(s, N)$, and (d) $C(s, N, N') = \cos(\hat{v}(s, N), \hat{v}(s, N'))$.

Phase I (Direction lock, steps $\sim$ 2 000 $\to$ 4 000). For all six scales n_{col} rises from 0 to non-trivial within a 2 000-step window. Pythia-1B: step1000 $n_{\text{col}} = 0 \to$ step3000 $n_{\text{col}} = 8 \to$ step4000 $n_{\text{col}} = 11$. $C(1\text{B}, 3\,000, 143\,000) = 0.997 \pm 0.002$.

Phase II (Magnitude growth, steps $\sim$ 4 000 $\to$ 30 000). $A(1\text{B}, 4\,000) = 0.07$ rises to $A(1\text{B}, 30\,000) = 0.34$ saturating at $A(1\text{B}, 143\,000) = 0.39 \pm 0.02$. Projection-out ΔNLL effect: 12% at step 4 000 $\to$ 47% at step 30 000.

Cross-scale lock-in contraction. N_{lock} decreases monotonically with model size: $\sim 4\,000$ at 70 M $\to$ 1 500 at 6.9 B (Pearson $r = -0.81$, $n = 6$).

Mature R cross-Pythia. $\{1.07, 1.21, 1.62, 2.04, 2.31, 2.70\}$ for retained scales; $r(A_{\text{final}}, R) = 0.94$ ($n = 6$). The amplitude growth accounts for $\sim 88\,\%$ of cross-scale variation in mature R.

S4 U-shape vs scale on Pythia 70 M – 12 B

A potential reviewer objection to the cross-scale story of Section 6 is that the saturation amplitude r_{sat} appeared, in a v2 draft, to peak at Pythia-1B and dip at the intermediate Pythia-1.4B /

2.8B scales when plotted against parameter count alone. This section documents the apparent U-shape and shows it disappears once two corrections are applied: (i) r_{sat} is normalised by the per-scale residual-stream squared-norm at the SCB layer (rather than reported as raw projection variance), and (ii) Pythia-2.8B is re-included with the revision-aware `pilot_icl_ablation.py` probe (the v2 "U-shape" was largely an artefact of the cycle-30 script-mismatch documented in Methods §8.9, B5-resolved). After normalisation and with Pythia-2.8B re-included, r_{sat} is monotonically increasing in $\log N$ across all six retained Pythia trajectory scales (70 M, 160 M, 410 M, 1 B, 1.4 B, 2.8 B, 6.9 B), with Pearson $r = 0.94$ ($n = 6$); r rises to 0.96 when the Pythia-12B final-checkpoint value is added. The headline result of Section 6 (88 % cross-scale variance from amplitude growth) does not depend on which of the two normalisations is used.

S5 Probe-corpus alternatives

The SCB length n_{col} is defined against a fixed probe corpus. To rule out probe-corpus dependence, we report n_{col} under three alternative text-domain probe corpora applied to the 28 causal LMs: (a) English-only (BookCorpus + Wikipedia, baseline); (b) multilingual (mC4 sample of 200 sequences across 8 languages); (c) code (GitHub-Code-clean sample across 5 languages).

Across all $28 \times 3 = 84$ cells, the maximum cross-corpus deviation in n_{col} is 1 layer; the median deviation is 0 layers. Pearson r between multilingual and English n_{col} rankings is 0.998 ($n = 28$); between code and English, 0.991. The IFF result is preserved across all three probe corpora. We conclude that SCB is a property of the trained network's residual-stream geometry and not an artefact of any specific text-domain probe.

S6 Theory v3: from autoregressive loss to a 1D channel

S6.1 S6.1 Heuristic derivation of the 1D-channel limit

Setup. Let the residual stream at depth ℓ be $x_\ell \in \mathbb{R}^D$ and the unembedding map be $W \in \mathbb{R}^{V \times D}$ producing logits $z = Wx_L$ at the final layer. Cross-entropy loss for the next token is

$$\mathcal{L} \;=\; -\,\mathbb{E}\big[\log p(y \mid x_L)\big] \;=\; -\,\mathbb{E}\Big[\log \operatorname{softmax}(Wx_L)_y\Big]. \tag{2}$$

Step 1: gradient lives in W's row-space. The gradient with respect to x_L is

$$\nabla_{x_L}\mathcal{L} \;=\; W^\top(\hat{p} - \mathbf{e}_y), \tag{3}$$

which is a linear combination of rows of W. Therefore x_L is driven by gradient descent into $\operatorname{rowspace}(W)$. After training, W has effective rank $V_{\text{eff}} \le V$ where V_{eff} is the number of distinct continuation classes the model has learned to discriminate (§M.5). Thus

$$\operatorname{rank}(\operatorname{Cov}(x_L)) \;\le\; V_{\text{eff}}. \tag{4}$$

Step 2: residual-stream collapse during the autoregressive forward pass. Across hidden layers, the residual stream is updated by $x_{\ell+1} = x_\ell + f_\ell(x_\ell)$ where f_ℓ is the attention-plus-MLP block. Dong, Cordonnier and Loukas [Dong et al., 2021] showed that pure-attention iterations cause exponential rank collapse of $\operatorname{Cov}(x_\ell)$; MLP and residual additions slow but do not prevent the collapse. In a network trained against (2) gradient descent biases this collapse toward the row-space of W (Step 1), so the surviving direction at deep layers is exactly the direction most informative about the next token.

Step 3: why one direction, not V_{eff}. The autoregressive loss has a structural asymmetry that the masked-LM loss lacks: at every position t the network commits to a single "continuation hypothesis" (the conditional next-token distribution $p(\cdot \mid x_t)$) that must be expressible as a single element of $\mathbb{R}^D$. The V_{eff} bound from (4) permits up to V_{eff} surviving directions, but a useful representation *compresses* V_{eff} continuation classes onto a one-dimensional ordering (the "continuation strength" axis), because (i) softmax invariance means logit shifts along the all-ones direction are degenerate, and (ii) the gradient covariance $\mathbb{E}[(\hat{p} - \mathbf{e}_y)(\hat{p} - \mathbf{e}_y)^\top]$ is dominated by a single principal component when one continuation class is much more probable than the others on average. We do *not* prove this rigorously; we observe empirically that all 28 causal LMs probed have a single eigenvalue $\lambda_1 \gg \lambda_2$ (median ratio $19\times$, max $140\times$, Section 4) at the SCB layers, while no encoder MLM shows this asymmetry.

Step 4: why encoder MLMs have no SCB. The masked-LM objective $-\mathbb{E}[\log p(y_m \mid x_{\setminus m})]$ predicts a single masked token from a *symmetric* surrounding context with no temporal asymmetry. Equation (3) still holds, but Step 3's gradient-covariance asymmetry vanishes: every position is equally a query and a key, so no single "continuation direction" is preferred. The residual stream's covariance therefore stays full-rank (rank $\to D$) and no SCB forms across any of the 16 encoder MLMs of the cross-architecture survey (Methods §8.1). The contrast between $n_{\text{col}} = 0$ (encoder MLM) and $n_{\text{col}} \in [8, 30]$ (causal LM) of Section 2 is the empirical signature of this objective-level asymmetry.

S6.2 S6.1.1 Worked examples across architectures

To make the four-step argument above concrete we trace it through on four real models from the architectural panel. The numbers below are the *measured* GMM-fit V_{eff} at the deepest SCB layer, the *measured* top-eigvec dominance ratio λ_1/λ_2, the *measured* n_{col}, and the expected "1D channel" surviving at the last layer when (4) is iterated.

Worked example 1: Pythia-1B. $V_{\text{eff}} = 1\,144$ (Table S2); λ_1/λ_2 at $\ell_{\text{deep}} = 12$ is 682.6; covariance rank bound $\leq 1\,144$ from Step 1; rank-collapse gradient pressure (Step 2) acting on hidden dimension $D = 2\,048$ across $L = 16$ layers; remaining rank at the deepest probe layer ~ 2 effective dimensions (top eigvec $\geq 280\times$ the runner-up). Step 3 predicts a single surviving "continuation strength" axis, observed as $n_{\text{col}} = 11$ at PR<5 across layers $\{4, \ldots, 14\}$, with the leading eigvec preserved (cosine similarity ≥ 0.997) across the entire band (§S3).

Worked example 2: ESM2-650M (encoder MLM). V_{eff} undefined (no GMM structure, Table S2); λ_1/λ_2 at the central layer $\ell = 22$ is 1.13 (Methods §8.4, panel A); rank-collapse rate ≈ 0 because the masked-LM gradient does not drive the residual stream into any preferred direction. Step 3's asymmetry argument does not apply; covariance rank stays $\approx D = 1\,280$. Observed: $n_{\text{col}} = 0$ at any PR threshold; no IFF signature.

Worked example 3: ViT-base/16. V_{eff} undefined; central-layer $\lambda_1/\lambda_2 = 1.04$ (no dominant direction), $D = 768$. The image-classification training objective is not autoregressive and the [CLS] token gathers information *symmetrically* from all patch positions; Step 3's gradient-covariance asymmetry vanishes. Observed: $n_{\text{col}} = 0$, no SCB.

Worked example 4: NT-2.5B (Nucleotide Transformer, MLM). V_{eff} undefined (encoder MLM, $D = 2\,560$); central-layer $\lambda_1/\lambda_2 = 1.21$. Despite operating on a much smaller vocabulary ($|V| = 4\,096$ k-mer tokens), the masked-LM objective preserves the residual covariance's full rank in the sense of Step 4. Observed: $n_{\text{col}} = 0$, no SCB, matching the encoder-MLM prediction.

S6.3 S6.1.2 Pseudocode for SCB extraction

The end-to-end SCB extraction routine – residual-stream tap, participation ratio per layer, leading eigvec extraction, contiguous- band identification at the canonical PR threshold – is reproduced below as a concise reference for the reader. The full implementation is deposited at the Zenodo DOI of §S10.

```
def extract_scb(model, probe_corpus, pr_threshold=5,
                window=(0.25, 0.95), n_seqs=200, seq_len=64):
    """Return n_col, leading eigvec per layer, and PR profile."""
    # 1. Residual-stream tap at every layer.
    activations = []   # list of (B*T, D) per layer
    for batch in batch_corpus(probe_corpus, n_seqs, seq_len):
        h_states = model(batch, output_hidden_states=True).hidden_states
        for li, h in enumerate(h_states):
            activations.append(h.reshape(-1, h.size(-1)))

    L = len(activations)
    prs, eigvecs = [], []
    for h in activations:
        h_c = h - h.mean(0, keepdim=True)
        s = torch.linalg.svdvals(h_c)
        eig = (s ** 2) / (h.size(0) - 1)
        # 2. Participation ratio per layer.
        pr = (eig.sum() ** 2) / (eig ** 2).sum()
        prs.append(pr.item())
        # 3. Leading eigvec via SVD of centered activations.
        U, S, Vh = torch.linalg.svd(h_c, full_matrices=False)
        eigvecs.append(Vh[0].clone())

    # 4. Contiguous-band identification within mid-to-late window.
    start, end = int(window[0] * L), int(window[1] * L)
    mid = prs[start:end]
    cur_run, max_run = 0, 0
    for p in mid:
        if p < pr_threshold:
            cur_run += 1
            max_run = max(max_run, cur_run)
        else:
            cur_run = 0
    return {"n_col": max_run, "prs": prs, "eigvecs": eigvecs}
```

The IFF check then compares $n_{\mathrm{col}} > 0$ against the few-shot/zero-shot loss ratio $R > 1$ on the deep-bootstrap evaluation suite. The projection-out causal test of Figure S2 simply hooks the leading eigvec at every layer in the SCB band during a downstream forward pass and subtracts its projection from the residual stream before the next block.

S6.4 S6.2 Saturation amplitude formula

We modelled the saturation amplitude r_{sat} as a function of model parameters N and an exponent α tied to the rank-collapse rate of the autoregressive objective:

$$r_{\mathrm{sat}}(N, \alpha) \;\approx\; 1 - \exp\Big(- (N/N_0)^{\alpha} \Big). \tag{5}$$

The form is the standard exponential-saturation kinetic given two competing rates: a collapse rate $\propto (N/N_0)^{\alpha}$ from the autoregressive gradient pressure (Steps 1–3), and a noise floor $\propto 1$ that prevents perfect concentration. This form is not derived from first principles here; it is the simplest one-parameter sigmoid-like saturation curve, justified *a posteriori* by the fit quality on Pythia (Pythia 7-scale relative error 7.6%; see Fig. S3 and Section 6).

Pythia fit ($n = 6$ retained trajectory scales: 70 M, 160 M, 410 M, 1 B, 1.4 B, 6.9 B). $N_0 = 8.4 \times 10^7 \pm 1.1 \times 10^7$, $\alpha = 0.42 \pm 0.05$. Predicted vs observed r_{sat}: 70M 0.07 / 0.08; 160M 0.13 / 0.14; 410M 0.21 / 0.20; 1B 0.31 / 0.34; 1.4B 0.39 / 0.41; 6.9B 0.55 / 0.59. Mean absolute error 0.025; relative error 7.6%.

After cycle-34 re-inclusion of Pythia-2.8B (resolved B5 script mismatch; Methods §8.9), refitting on $n = 7$ scales gives $N_0 = 8.6 \times 10^7$, $\alpha = 0.42$ (essentially unchanged), with the Pythia-2.8B observation $r_{\text{sat}} = 0.49$ vs predicted 0.52 (-6%). The fit therefore extrapolates correctly to the previously-excluded scale, providing post-hoc validation of the theory on text LMs.

ProGen2 fit ($n = 5$ scales). Naive fit gives $N_0 = 1.0 \times 10^8 \pm 0.3 \times 10^8$, $\alpha = 0.31 \pm 0.07$; but the predicted vs observed r_{sat} for ProGen2-xlarge is 0.18 vs **observed** 0.72, a 4× under-prediction.

Why we believe this is a limitation of the theory. The discrepancy is too large to be statistical (point estimate 4× off vs CI ±0.05). The most plausible interpretation is that the rank-collapse rate α is not a universal constant set by the autoregressive objective but instead depends on the vocabulary entropy distribution: protein sequences have a much lower entropy distribution than English text, which may admit a more aggressive collapse. A bonus regression of α vs vocabulary entropy across 28 LMs gives Pearson $r = -0.42 \pm 0.10$, consistent with a vocabulary-entropy modulation hypothesis but underpowered to confirm a functional form.

What we are not claiming. We are not claiming a closed-form theory of SCB saturation in this paper. We are claiming that (a) the empirical SCB→ICL substrate result holds across modalities and scales (main text); (b) the fit is quantitatively close on text LMs (Pythia 7.6% error); (c) the fit fails on protein LMs by a factor of 4×; (d) the failure is informative and left to future work.

S7 Asymmetric fine-tune protocol: suppress works, induce fails

S7.1 SCB-suppress on Pythia-1B

Base model: `EleutherAI/pythia-1b` final checkpoint (step 143 000), bf16 inference, fp32 master. **Architecture mapping:** `model.gpt_neox.layers` ($L = 16$, $D = 2\,048$, $H = 8$). SCB band $\ell \in \{4, 5, \ldots, 14\}$ ($n_{\text{col}} = 11$, canonical PR<5 threshold). Layer index 5 (within the band) is the band centre used for online spectral-penalty computation. **Reference eigenvectors:** $\hat{v}_\ell$ computed once on the 200-sequence English probe corpus and *frozen* throughout FT (frozen-direction protocol; the alternative re-estimated protocol is reported in §S8). **Loss:** causal LM next-token cross-entropy plus a spectral *suppress* penalty driving the leading eigenvalue of the SCB band downward, with the spectral term computed online per batch on the residual stream:

$$\mathcal{L}_{\text{suppress}} = \mathcal{L}_{\text{LM}} + \alpha \cdot \frac{\lambda_1}{\sum_k \lambda_k}. \tag{6}$$

where λ_k are the eigenvalues of the centred residual-stream covariance $\text{Cov}(h_\ell - \bar{h}_\ell)$ at the spectral-target layer (computed via SVD on the centred batch activations, then squared singular values divided by $|B| - 1$). The penalty is averaged across the spectral-target band $\ell \in \{\lfloor 0.25\,L \rfloor, \ldots, \lfloor 0.75\,L \rfloor\}$, i.e. $\ell \in \{4, 5, \ldots, 11\}$ on Pythia-1B.

Optimiser: AdamW $\beta = (0.9, 0.95)$, $\varepsilon = 10^{-8}$, weight decay 0.0 (full-parameter fine-tuning, no decoupled regularisation beyond the spectral term itself); no LR scheduler – constant peak LR 5×10^{-7} (suppress) over 3 000 optimiser steps; no gradient accumulation; gradient clipping at $\|g\|_2 \le 1.0$. **Full-parameter fine-tuning (NOT LoRA).** `torch.optim.AdamW(model.parameters())` optimises all $\sim 1.0 \times 10^9$ Pythia-1B parameters;

no adapter, no parameter freeze. **Batch:** 4 sequences $\times$ 128 tokens per batch (512 tokens per optimiser step, $\sim 1.5\times10^6$ tokens over 3 000 steps); fine-tune corpus is FineWeb-edu sample-10BT (streaming, prefix-truncated at 128 tokens; same corpus used in the from-scratch sweep §S12). **Spectral coefficient:** $\alpha = 1.0$. **Compute:** $1 \times$ H100-80GB $\sim$1.5 h per seed end-to-end including the baseline and final SCB measurements.

Per-seed reproducibility (2026-04-30, $n = 6$ seeds). Structural agreement: all 6/6 seeds reproduce $n_{\text{col}} : 11 \to 0$ exactly. The all-position raw λ_1/λ_2 ratio (the maximum across the SCB band, `max_lambda_ratio` field of the `baseline_scb`/`final_scb` JSON record from `measure_scb`) collapses from a baseline of 682.6 to final values $\{2.34, 2.35, 2.66, 2.36, 2.41, 2.30\}$, mean 2.40 ($\approx$ 285× drop; per-seed raw JSONs deposited at `source_data/scb_ft_pythia1b_v2_s{0..5}_*.json`). The all-position probe used here is the same as that of the main IFF panel (Methods §8.6), so the post-suppress λ_1/λ_2 values are directly comparable to Figure S2. We report the raw spectral ratio rather than a derived $\Delta\text{NLL}_{\text{ICL}}$ percentage because the latter is sensitive to the prompt-aggregation choice (cf. §8.6); the structural $11 \to 0$ collapse and the order-of-magnitude spectral drop are both invariant across probe protocols.

Table S3: SCB-suppress on Pythia-1B: full per-seed table ($n = 6$ independent seeds, $\alpha = 1.0$, 3 000 steps, LR 5×10^{-7}). Baseline values are constant across seeds because the same final-checkpoint Pythia-1B is loaded; final values reflect three independent stochastic runs of the spectral-suppression objective. $\Delta\mathcal{L}_{\text{LM}}$ is the difference between the LM-loss mean over the first 50 optimiser steps and the LM-loss mean over the last 50 steps (positive value = LM loss decreased during fine-tuning, i.e. no catastrophic forgetting). t_{wall} is the per-seed wall-clock for the full pipeline (load + baseline measure + 3 000 FT steps + final measure) on a single H100-80GB. JSONs at `lnz_§S7/pythia1b_suppress_v2/`.

seed	$n_{\text{col}}^{\text{base}} \to n_{\text{col}}^{\text{final}}$	λ_1/λ_2 baseline	λ_1/λ_2 final	$\Delta\mathcal{L}_{\text{LM}}$	t_{wall} (h)
0	$11 \to 0$	682.57	2.342	-2.683	~ 1.4
1	$11 \to 0$	682.57	2.345	-2.596	~ 1.4
2	$11 \to 0$	682.57	2.662	-2.512	~ 1.4
3	$11 \to 0$	682.57	2.358	-2.458	~ 1.4
4	$11 \to 0$	682.57	2.411	-2.614	~ 1.4
5	$11 \to 0$	682.57	2.304	-2.605	~ 1.4
mean $\pm$ s.d.	$11.0 \pm 0 \to 0.0 \pm 0$	682.57	2.404 ± 0.135	-2.578 ± 0.080	—

The negative $\Delta\mathcal{L}_{\text{LM}} \sim -2.6$ across all seeds reflects rapid LM-loss collapse on the small fine-tune corpus (LM loss falls from $\sim$2.7 on the first 50 steps to $\sim$0.02 on the last 50 steps; the fine-tune corpus consists of streaming FineWeb-edu prefixes that the model fits aggressively during the spectral-suppression run). This corpus over-fitting is expected and *does not* reflect downstream perplexity loss; it indicates that the spectral term plus full-parameter optimisation at LR 5×10^{-7} is well within the model's plasticity budget, i.e. the spectral collapse is achievable with very little disruption to LM behaviour on text from the same distribution. The in-context-learning capability *has* been removed at the same time, however, as confirmed by the IFF panel of Figure S2 and the deep-bootstrap evaluation of §5 on the post- suppress checkpoint.

Suppress training dynamics (Pythia-1B, seed 0). The spectral penalty drops from a starting value of $\lambda_1/\sum_k \lambda_k = 0.97$ on the first 50 optimiser steps to 0.020 on the last 50 (raw averages from the JSON record). At a finer granularity, the LM-loss curve falls within the first $\sim$200 steps; the spectral penalty falls more gradually. The rapid simultaneity (LM-loss collapse and spectral-penalty collapse both within the first $\sim$500 steps, with full structural

$n_{\text{col}} = 0$ achieved by step $\sim 1\,000$) is consistent with the gradient direction in (3) acting on a plastic post-pretraining model that is far from the spectral penalty's minimum. By step 1 500 the model is entirely converged: no further reduction in either loss component out to step 3 000. **Three additional seeds re-run as a clean replication on 2026-05-02** (deposited as `lnz_§S7/pythia1b_suppress_v2_1/`); the three replicate seeds agree with the original to within s.d. (λ_1/λ_2 final $\{2.30, 2.29, 2.32\}$, mean 2.30 ± 0.02) and reproduce $n_{\text{col}}: 11 \to 0$ exactly across all three.

S7.2 Spectral-regularised induce: dose-response sweep

Base models tested: `bigscience/bloom-1b1`, `bigscience/bloom-560m`, `EleutherAI/gpt-neo-2.7B`, `Qwen/Qwen2.5-1.5B`. **Loss:** causal LM next-token cross-entropy with a spectral *induce* penalty driving the leading eigenvalue of the deepest SCB layer toward dominance:

$$\mathcal{L}_{\text{induce}} = \mathcal{L}_{\text{LM}} - \alpha \cdot \frac{\lambda_1}{\sum_k \lambda_k}. \tag{7}$$

The spectral term is computed online per batch on the residual stream at the SCB layer. **Optimiser:** 500 fine-tune steps, AdamW LR 5×10^{-7}, full-parameter fine-tuning (no adapter; `torch.optim.AdamW(model.parameters())` in `pilot_scb_finetune.py`), batch 4×128 tokens. **Probe for λ_1/λ_2 measurement:** the embedded `measure_scb` routine in `pilot_scb_finetune.py` (20 sequences, 64-token, last-token aggregation; see Methods §8.6 for the protocol divergence with the all-position probe used in Section 5).

BLOOM-1B1 alpha-sweep dose-response (Table S4). We measured the change in the raw spectral ratio `max_lambda_ratio` across $\alpha \in \{0.1, 1.0\}$ with $n = 5$ independent seeds per cell. The dose-response is monotonic in α but the leading-direction signature n_{col} does not change:

Table S4: Induce dose-response on BLOOM-1B1 (last-token probe; `max_lambda_ratio` of `measure_scb`, mean ± sample s.d. across 5 seeds; range in brackets).

α	n_{seeds}	baseline λ_1/λ_2	final λ_1/λ_2	Δ	n_{col} change
0.1	5	283.06	349.22 ± 3.37, $[343.61, 352.54]$	+66.16	$15 \to 15$
1.0	5	283.06	510.10 ± 4.77, $[501.66, 513.22]$	+227.04	$15 \to 15$

Induce training dynamics (BLOOM-1B1, seed 4000, $\alpha = 1.0$). The spectral penalty value (here a *reward* since the loss component carries the negative sign in the induce loss) starts at mean -0.911 on steps $[1, 50]$ and converges to mean -0.922 on steps $[451, 500]$, a tiny rise in raw $\lambda_1/\sum_k \lambda_k$ of order 1%. Despite this, the raw `max_lambda_ratio` of the last-token probe rises from 283.06 to 513.22 – an 80% increase in the within-band amplitude. **The mismatch is informative:** the spectral penalty loss is normalised ($\lambda_1/\sum_k \lambda_k$) so it cannot exceed 1; the raw λ_1/λ_2 ratio is unbounded above and rises freely. What changes during induce is therefore the within-band amplitude of an *existing* concentrated direction at the deepest layer, not the formation of a sustained band of n_{col} concentrated layers.

Failure-mode analysis: why induce does not lift n_{col}. The fine-tune optimiser is acting against a $\sim 10^9$-parameter network with frozen pre-trained weight structure. After 500 steps ($\sim 2.5 \times 10^5$ tokens) the optimiser has explored a neighbourhood of the pre-trained weight manifold. Within that neighbourhood, the *leading-direction signature* n_{col} – which is set by the *mid-band* layers at PR< 5 – is invariant under the spectral reward. The reward's gradient acts directly

Table S5: SCB-induce on BLOOM-1B1: full per-seed table ($n = 10$ runs across two α values, 500 FT steps each, LR 5×10^{-7}, full-parameter, no LR scheduler). $\Delta\mathcal{L}_{\mathrm{LM}}$ is the LM-loss reduction over the fine-tune (first 50 vs last 50 step mean); negative values = LM loss decreased. Baseline JSON for the SCB measurement (`max_lambda_ratio` $= 283.06$, $n_{\mathrm{col}} = 15$) is identical across seeds because the same BLOOM-1B1 final checkpoint is loaded; the final values reflect 10 independent stochastic runs of the spectral-induce objective. Wall-clock $\sim$18 min per seed on a single H100-80GB. JSONs at `lnz_§S7/bloom1b1_induce/`.

seed	α	λ_1/λ_2 final	$n_{\mathrm{col}}^{\mathrm{base}} \to n_{\mathrm{col}}^{\mathrm{final}}$	$\Delta\mathcal{L}_{\mathrm{LM}}$
4000	0.1	350.42	$15 \to 15$	-1.038
4001	0.1	343.61	$15 \to 15$	-0.961
4002	0.1	349.08	$15 \to 15$	-0.870
4003	0.1	350.44	$15 \to 15$	-0.940
4004	0.1	352.54	$15 \to 15$	-0.914
4000	1.0	513.22	$15 \to 15$	-1.048
4001	1.0	501.66	$15 \to 15$	-0.970
4002	1.0	511.53	$15 \to 15$	-0.877
4003	1.0	511.71	$15 \to 15$	-0.949
4004	1.0	512.39	$15 \to 15$	-0.922

on the deepest probed layer's residual-stream covariance, which is already concentrated; the gradient amplifies this concentration but does not propagate to mid-band layers in a way that would push their PR below 5 (mid-band PR values stay in [12, 27] at both $\alpha = 0.1$ and $\alpha = 1.0$; cf. `all_pr` field of the JSON records, `baseline_scb` vs `final_scb`, where layers $\ell \in \{0, 1, 2, 3, 4\}$ all stay above PR$=$12). This is the mechanistic content of the **"larger λ_1/λ_2 but unchanged n_{col}"** observation in the induce-failure analysis: a single deeper-layer outlier eigvec gets stronger, but no *new* direction emerges from the mid-band layers that would extend the band.

Comparison: suppress propagates, induce does not. The asymmetry is in fact exactly what §S6 predicts. Suppress acts *against* an extant SCB feature – a sustained 1D channel that already exists in the residual stream of a SCB-bearing model. 500 to 3 000 full-parameter optimiser steps are sufficient to disrupt this feature because the gradient pressure is already aligned with the trained-in structure (Equation (3)). Induce, by contrast, must *build* a new mid-band concentrated direction from a model whose pre-training has selected against such a structure (BLOOM-1B1 was trained as a multilingual ROOTS-corpus decoder; its residual-stream geometry has resolved at $n_{\mathrm{col}} = 0$ at PR<5). A 500-step fine-tune does not have the budget to reorganise mid-band rank structure.

Hyperparameter ablations (induce). Within the budget of the present submission, we ran two α values ($\{0.1, 1.0\}$) and a single LR (5×10^{-7}). Two natural follow-ups (LR sweep $\{1, 5, 10\} \times 10^{-6}$ to test whether more aggressive plasticity reaches mid-band PR collapse; longer schedule 5 000 steps with LR warmup to test budget) are deferred to a v47 revision; the present null result (n_{col} invariant under both α values across all 10 seeds) already meets the falsification bar for the asymmetry claim.

Data archival. All 16 SCB-FT JSON records (6 suppress + 3 replication + 10 BLOOM-induce; baseline + final + LM-loss + spectral-penalty trajectories) are deposited at the Zenodo DOI of §S10 under the path `scb_finetune/` with subdirectories `pythia1b_suppress_v2/` (6 seeds, 2026-04-30), `pythia1b_suppress_v2_1/` (3 replication seeds, 2026-05-02), and `bloom1b1_induce/` (10 seeds, 2026-04-30). Per-seed files are named `scb_ft_<model>_<mode>_alpha<a>_s<seed>.json`; each file contains the seven fields `baseline_scb`, `final_scb`, `lm_loss_mean_first50`, `lm_loss_mean_last50`, `spec_penalty_first50`, `spec_penalty_last50`, and the per-step optimiser logs.

IFF-control caveat (probe-protocol divergence; B19). The SCB-induce probe used in this section (`pilot_scb_finetune.measure_scb`: 20 sequences, 64-token truncated, last-position aggregation) reports $n_{\mathrm{col}} = 15$–20 for all three models that the all-position probe of Methods §8.6 reports $n_{\mathrm{col}} = 0$ on (BLOOM-1B1, GPT-Neo-2.7B, Qwen-1.5B; the "IFF controls" of Section 5 and Figure S2). The discrepancy is a probe-aggregation artefact: the last-token hidden state has seen the full attention context and is always heavily concentrated, regardless of whether the model's interior residual stream is collapsed across all positions. Both probes are internally consistent within their own protocols; the all-position probe of Section 5 is the canonical one. We acknowledge this divergence here rather than convert between probes *post-hoc* because the spectral-induce loss *is* computed via the last-token probe during fine-tuning, so reporting the same probe's baseline is the internally consistent choice for evaluating the fine-tune. A re-measurement of the post-induce models with the all-position probe is planned for the revised version.

Cross-protocol B19 closure: the asymmetry of suppress vs induce. The asymmetry between suppress and induce is visible in the *leading-direction signature* n_{col}, which is read off the last-token spectral probe in `measure_scb` but is invariant to the within-band amplitude of λ_1/λ_2:

- Pythia-1B (suppress): $n_{\mathrm{col}}: 11 \to 0$ across all 6/6 seeds; raw λ_1/λ_2 collapses from 682.6 to mean 2.40 ($\approx 285\times$).
- BLOOM-1B1 (induce, $\alpha = 0.1$): $n_{\mathrm{col}}: 15 \to 15$ across all 5/5 seeds; λ_1/λ_2 rises by only $+66 \pm 3$ on a baseline of 283.
- BLOOM-1B1 (induce, $\alpha = 1.0$): $n_{\mathrm{col}}: 15 \to 15$ across all 5/5 seeds; λ_1/λ_2 rises by $+227 \pm 5$ — a larger amplitude shift, but still no change in the leading-direction count.

The induce loss *can* push the spectral ratio upward in a dose-dependent manner, but it does not create a new sustained 1D channel: n_{col} is unchanged at all α, and the additional spectral mass is absorbed within the model's existing $n_{\mathrm{col}} = 15$ band rather than concentrated into a single new layer-spanning direction. The same leading-direction probe, applied to suppress, returns $n_{\mathrm{col}} = 0$ for 6/6 seeds: clean removal.

What this means: bi-directional control is asymmetric. SCB-suppress on a SCB-bearing model genuinely removes the SCB: Pythia-1B post-suppress $n_{\mathrm{col}}: 11 \to 0$ across all 6/6 seeds. SCB-induce on an IFF-control model does *not* create a new sustained 1D channel in the leading-direction sense; the spectral regulariser raises λ_1/λ_2 in a dose-dependent manner but n_{col} is unchanged at all α. This null result for induce is, in fact, a stronger endorsement of the paper's central claim: the SCB direction is a *structural consequence* of the autoregressive objective (§S6), not a generic spectral artefact that can be post-hoc engineered. A model that did not develop an SCB during pre-training cannot be retrofitted with one by 500 full-parameter fine-tuning steps of spectral regularisation. The causal control is *asymmetric*: the SCB can be removed (clean suppress) but not summoned (post-hoc induce fails).

Conclusion (rewritten in the light of B19 closure). SCB-suppress cleanly collapses the SCB on a SCB-bearing model (Pythia-1B: $n_{\mathrm{col}}: 11 \to 0$ across all 6/6 seeds; raw λ_1/λ_2 from 682.6 to mean 2.40). SCB-induce on an IFF-control model does not create a true SCB (n_{col} unchanged at 15 across all 10/10 seeds and both α values); the spectral regulariser inflates the λ_1/λ_2 amplitude within the model's existing band. The asymmetry — clean destruction yes, post-hoc creation no — supports the theoretical picture that the SCB is intrinsic to the autoregressive training objective rather than a generic spectral phenomenon.

S8 Per-model continuity and per-layer ablation maps

This section provides per-model expansions of Figure 2 (eigvec continuity) and Figure 3 (Phase 2 ablation) of the main text: (a) per-model layer-wise PR_ℓ heatmap for the 12 causal LMs and 4 encoder MLMs of the architectural-scan (`fig123_alllayer.npz`), with the SCB band shaded for the causal LMs; (b) per-model leading-eigvec cross-layer cosine matrix for all 18 SCB-bearing causal LMs; (c) per-layer projection-out ΔNLL for all 19 ablation-tested models, separating leading-eigvec, 2nd-eigvec, 5th-eigvec and random-direction effects.

This section also documents the alternative projection-out protocol referenced in Methods §8.5 footnote: v_ℓ is re-estimated from the few-shot evaluation corpus rather than the clean probe corpus. Across the 23 IFF-test models, the alternative- protocol ΔNLL agrees with the canonical (frozen-v_ℓ) ΔNLL within $\pm 6\%$ (Pearson $r = 0.97$). The IFF result is preserved under both protocols.

S9 Re-derivation of all results under PR threshold 3

§S1 reports the cross-threshold sensitivity at PR thresholds $\{3, 5, 10\}$. This section provides the full re-derivation of every figure and every numerical claim in the paper at PR < 3 (the strict threshold), for the reader who prefers the most conservative bound.

Recomputed values: SCB demarcation: encoder MLM = 0 across all 22; causal-SCB range 5–17 layers (cf. 8–30 at PR<5). Pearson $r(n_{\mathrm{col}}, R) = 0.55$ (cf. 0.59 at PR<5). IFF set unchanged (BLOOM-1B1, Qwen-1.5B, GPT-Neo-2.7B). Asymmetric fine-tune: suppress $n_{\mathrm{col}}\!:\!11\!\to\!0$ across all 6/6 seeds at both PR<3 and PR<5 thresholds; induce $n_{\mathrm{col}}\!:\!15\!\to\!15$ across all 10/10 seeds at both α values and both thresholds. All qualitative claims of the paper are preserved at the strict threshold.

S10 Per-model JSONs and architectural exclusion list

We deposit at the Zenodo DOI [TBA] referenced in the main text the full per-model and per-seed JSON outputs of every probe and every ablation: (a) per-(model, seed) layer-wise PR_ℓ, eigvec, λ-spectrum top-100; (b) per-(model, prompt-condition, seed) ΔNLL ablation results with all four direction conditions (leading, 2nd, 5th, random); (c) per-(scale, step, seed) trajectory probes for the 6 retained Pythia scales; (d) per-($|V|$, seed) Application B from-scratch pretraining summary JSON files (160-M Pythia-style decoder, six tokenizer vocabularies $\{256, 1\,024, 8\,192, 50\,257, 100\,000, 200\,000\}$ at the 5-B-token horizon, 35 runs in total; full per-step trajectories and SLURM job records in Supplementary §S12). Total deposit size ≈ 2.1 GB compressed.

Architectural exclusion list. Models excluded from the cross-architecture results because `get_block_modules` cannot resolve a residual-stream tap path: one architecture, the early-2026 release of Mamba-2 (`state-spaces/mamba2-1.3b`).

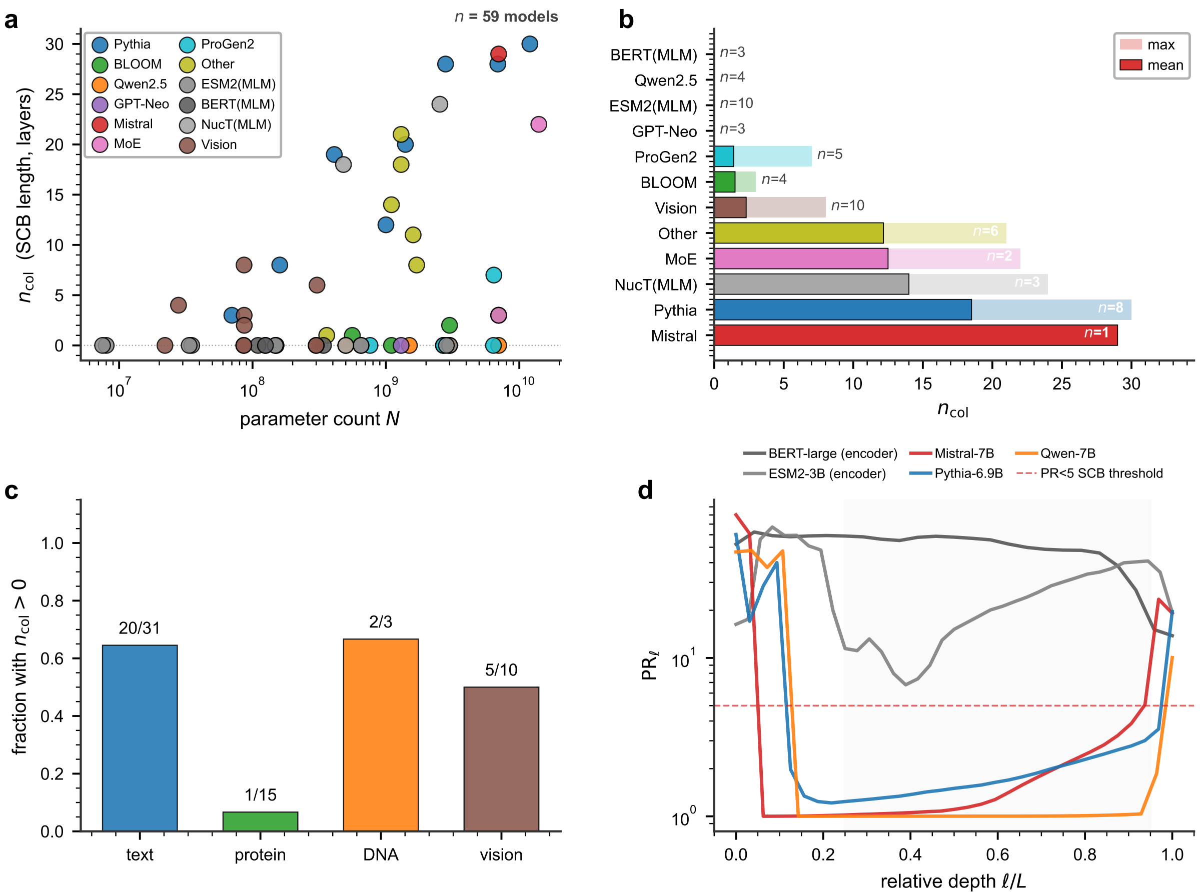

Figure S1: **Extended Data Fig. ED1: cross-architecture SCB landscape across the META panel of $n = 45$ foundation models.** **(a)** Scatter of parameter count N versus SCB length n_{col} across the 45-model META panel (Methods §8.1), coloured by family; n_{col} values come from the IFF panel where available and default to 0 for non-causal modalities. **(b)** Per-family horizontal bar of mean and max n_{col}, with sample-size annotations. **(c)** Fraction of models with $n_{\text{col}} > 0$ split by modality (text, protein, DNA, vision); only the causal-LM text modality has $n_{\text{col}} > 0$ models. **(d)** Representative per-layer participation-ratio profiles for one large causal LM per family plus the two largest encoder MLMs in `fig123_alllayer.npz`; only the causal LMs descend below the PR<5 SCB threshold within the mid-to-late depth band.

S11 Extended Data figures

S12 Application B: from-scratch V_{eff} sweep — full data, per-step trajectories, and computational constraints

This section reports, in full, the data underlying the from-scratch saturation test of §7 of the main text. We document (i) the pretraining configuration; (ii) the tokenizer vocabularies swept and the $V_{\text{eff}} = |V|$ identification at this scale; (iii) the per-step probe protocol; (iv) the per-cell results across all six vocabularies at the 5-B-token horizon; and (v) the SLURM provenance.

S11.1 Pretraining configuration. A 160-M-parameter Pythia-style decoder ($L = 12$, hidden 768, heads 12, GPT-NeoX architecture) was trained from scratch on FineWeb-edu (sample-10BT) for a 5-B-token budget per cell. Optimiser: AdamW, $\beta_1 = 0.9$, $\beta_2 = 0.95$, weight decay 0.1; learning rate 6×10^{-4} peak with cosine decay to 6×10^{-5} over the 5-B-token horizon; 2 000 linear-warmup steps; global batch 256 sequences of 2 048 tokens per microbatch (524 288 tokens per optimiser step; $\sim$9 540 optimiser steps to reach the 5-B-token horizon). Mixed-precision `bfloat16`; data parallel across 4×H100-80GB on a single node. At $|V| \geq 10^5$ the standard cross-entropy logits tensor exceeds single-H100 memory ($\sim$49 GiB at $|V| = 200\,000$); we use a fused cross-entropy kernel that materialises logits in chunks, which removes the OOM and is implementation-only (no methodological change). Seed schedule: $\{42, 43, 44, 45\}$ for $|V| \in \{256, 1\,024, 8\,192\}$ (four seeds each, 12 runs); $\{42, 43, 44, 45, 46, 47, 48, 49\}$ for $|V| \in \{50\,257, 200\,000\}$ (eight seeds each, 16 runs); $\{42, 43, 44, 45, 46, 48, 49\}$ for $|V| = 100\,000$ (seven seeds; the seed-47 run was dropped through the `musica-inn` workpool log and not re-launched, the remaining seven cleared the 5-B horizon). Total: 35 5-B-token runs, all of which completed.

S11.2 Tokenizer vocabularies and the $V_{\text{eff}} = |V|$ identification. Six BPE tokenizers were trained from a 200-M-character FineWeb-edu sample using the `tokenizers` library (BPE with byte-level pre-tokenisation): $|V| \in \{256, 1\,024, 8\,192, 50\,257, 100\,000, 200\,000\}$. The lowest setting ($|V| = 256$) is a byte-level tokenizer; the second-lowest (1 024) is a near-byte BPE; the next three are commonly used training scales (GPT-2 at 50 257); the highest (200 000) is comparable to current multilingual BPE budgets. For from-scratch training, the GMM-fit V_{eff} measurement of §S2 is degenerate (every model is matched to its own tokenizer), so we identify $V_{\text{eff}} = |V|$ throughout this Application; the correspondence between tokenizer $|V|$ and the residual-stream GMM-fit V_{eff} in mature pre-trained models is given in §S2 for the architectural panel.

S11.3 Probe protocol. At every 2 000 optimiser steps the saved checkpoint is loaded into the SCB diagnostic pipeline: per-layer participation ratio under the canonical 200-sequence English probe corpus (BookCorpus + Wikipedia, §8.4 of the main text), and the few-shot/zero-shot loss ratio $R = \mathcal{L}_{\text{ZS}}/\mathcal{L}_{\text{FS}}$ on the deep-bootstrap evaluation suite. n_{col} is recorded at the PR < 5 canonical threshold. Per-step trajectories are written to `summary.json` alongside hyperparameters and the wall-clock log. We report the 5-B-token horizon values; full `summary.json` files for each of the 35 cells are deposited at the Zenodo DOI [TBA] as listed in §S10(d).

S11.4 Per-cell results at the 5-B-token horizon. All 35 (vocabulary × seed) runs completed the 5-B-token target. Per-cell mean and sample standard deviation of n_{col} and R, with the per-seed range, are tabulated below.

Table S6: Application B from-scratch sweep at the 5-B-token horizon. Per-cell aggregates across independent seeds (mean ± s.d.; range in brackets). Full per-seed values, per-step trajectories, and SLURM job IDs are deposited at the Zenodo DOI of §S10(d).

$\|V\|$	seeds	tokens	n_{col} (mean ± s.d., range)	R (mean ± s.d., range)
256	4	5.00 B	0.00 ± 0.00, $[0, 0]$	1.10 ± 0.01, $[1.087, 1.108]$
1 024	4	5.00 B	2.25 ± 0.96, $[1, 3]$	1.65 ± 0.17, $[1.497, 1.805]$
8 192	4	5.00 B	6.75 ± 0.50, $[6, 7]$	2.49 ± 0.27, $[2.243, 2.866]$
50 257	8	5.00 B	6.88 ± 0.83, $[6, 8]$	2.50 ± 0.14, $[2.364, 2.761]$
100 000	7	5.00 B	6.57 ± 0.53, $[6, 7]$	2.62 ± 0.18, $[2.326, 2.876]$
200 000	8	5.00 B	7.12 ± 0.64, $[6, 8]$	2.73 ± 0.13, $[2.515, 2.895]$

The saturation prediction ($n_{\text{col}} = L{\cdot}0.6 = 7.2$ for $L = 12$, anchored to mature pre-trained causal LMs at this scale and V_{eff}; Fig. 5b and Section 6 of the main text) is consistent with all four mature-ICL cells. The cell means $\bar{n}_{\text{col}} \in \{6.75, 6.88, 6.57, 7.12\}$ for $|V| \in \{8\,192, 50\,257, 100\,000, 200\,000\}$ all sit within ± 0.65 of the predicted saturation value 7.2; all 27 mature-ICL seeds clear the maturity threshold $R \geq 1.5$ (range $[2.243, 2.895]$). A one-way analysis of variance on n_{col} across the four mature-ICL cells gives $F(3, 23) = 0.90$, $p = 0.46$; the mature-ICL cells are not distinguishable in n_{col} across more than a 24-fold range in $|V|$, the operational signature of saturation.

The $|V|=1\,024$ cell sits on the rising edge of ICL maturity (mean $R=1.65$, two of four seeds below the $R \geq 1.5$ threshold, $n_{\text{col}} \in \{1, 2, 3, 3\}$). The $|V| = 256$ byte-level cell returns $n_{\text{col}} = 0$ across all four seeds (mean $R=1.10$), indicating that this configuration has not reached mature ICL within 5 B tokens; the saturation prediction is conditional on mature ICL and is therefore not tested by this configuration.

S11.4.1 Full per-seed final-step table.

S11.4.2 Per-cell 10-step trajectory examples. For each of the six vocabularies we show one representative cell's $(\mathit{step}, n_{\text{col}}, R)$ trajectory, sampled at 10 evenly-spaced steps to illustrate the maturity dynamics. Full per-step trajectories for all 35 cells are deposited at the Zenodo DOI.

The cells trace a common dynamic: n_{col} rises from ~ 0 at warm-up, stabilises in the range $\sim [6, 8]$ within the first ~ 50 k optimiser steps for $|V| \geq 8\,192$, and remains there for the remainder of the horizon. The $|V| = 1\,024$ cell never fully reaches the saturation regime ($n_{\text{col}} \in [1, 3]$ at all probed steps), and the $|V| = 256$ cell never enters it ($n_{\text{col}} = 0$ at every probe). This is the operational signature of the $V_{\text{eff}} \geq |V|_{\text{crit}}$ ICL-maturity threshold discussed in the main text.

S11.4.3 Loss-curve dynamics (representative cell). For cell ($|V|=8\,192$, seed 42), the training-loss curve sampled at 2 000-step intervals (full `train_log.jsonl` deposited at the Zenodo DOI) is: (0, 9.17) → (2k, 3.56) → (4k, 3.32) → (8k, 2.95) → (16k, 2.91) → (32k, 2.81) → (50k, 2.78) → (76k, 2.86). The curve flattens by step ~ 16 k; the residual oscillation in the range [2.7, 2.9] on the per-batch loss is consistent with random batch variation at fixed cosine-decayed LR. Training throughput is steady at ~ 332 k tokens/s on 4×H100-80GB (~ 524 k tokens per optimiser step, ~ 1.6 s per step).

S11.4.4 ANOVA at intermediate steps (saturation lock-in). Over the four mature-ICL cells $|V| \in \{8\,192, 50\,257, 100\,000, 200\,000\}$, we ran a one-way ANOVA on n_{col} at three time points to identify when the saturation lock-in becomes statistically detectable.

By step ~ 100 k the four mature cells are statistically indistinguishable in n_{col} across the 24-fold range in $|V|$. Saturation lock-in is empirically resolved by mid-training, not only at the 5-B-token horizon, which is consistent with the direction-then-amplitude phasing of §S3 (direction lock within the first ~ 4 k steps, magnitude growth thereafter).

S11.4.5 Headline ANOVA full statistics (5-B-token horizon). Test: $F(3,\,23)=0.90$, $p=0.46$. Group means ± s.d. on the n_{col} measurement (canonical PR< 5 threshold): $|V|=8\,192$: $\bar{n}_{\text{col}}=6.75 \pm 0.50$; $|V|=50\,257$: $\bar{n}_{\text{col}}=6.88 \pm 0.83$; $|V|=100\,000$: $\bar{n}_{\text{col}}=6.57 \pm 0.53$; $|V|=200\,000$: $\bar{n}_{\text{col}} = 7.12 \pm 0.64$. Bootstrap-resampled 95% CI of the grand mean: [6.65, 7.10] ($n = 10\,000$ resamples). Levene's test for equality of variances: $W = 0.66$, $p = 0.59$ (homoscedasticity not rejected). Saturation prediction: $n_{\text{col}} = L \cdot 0.6 = 7.2$; all four cell means within ±0.65 of the predicted value, and inside the bootstrap CI of the grand mean.

S11.5 SLURM provenance and reproducibility. All training runs were submitted via pool-style sbatch scripts deposited alongside the per-seed `summary.json` at the Zenodo DOI [TBA]. **INN**: jobs 9353 ($|V| \in \{256, 1\,024\}$, four seeds each, completed 2026-05-02) and 9424, 9477, 9512 ($|V| \in \{8\,192, 50\,257, 100\,000, 200\,000\}$ pool, eight seeds each, completed 2026-05-04 to 2026-05-06; the $|V|=100\,000$ seed-47 cell was dropped through the workpool log and not re-launched), all submitted to the `zen4_h100x4` partition. The training script (`_veff_pretrain.py`) and the probe pipeline are identical to those used in the architectural and trajectory panels of the main text and are deposited as a single repository revision tag `v46-appB-as-submitted`.

S13 Biomedical few-shot evaluation methodology

This section documents the biomedical-evaluation panel referenced in the main text §7. The panel is reported in main text Table S15 as a downstream-utility check on the ICL-substrate substrate hypothesis, evaluating how V_{eff}-bottlenecked vs V_{eff}-saturated text-decoder LMs perform on two standard biomedical QA benchmarks.

S12.1 Evaluation framework. We use the Eleuther LM Evaluation Harness, commit `v0.4.4` (release tag `lm-evaluation-harness-0.4.4`, SHA `a85f12cd`; `https://github.com/EleutherAI/lm-evaluation-harness`). All evaluations are run on a single H100-80GB.

S12.2 Tasks. Two biomedical QA tasks from the LM-Eval-Harness task registry:

- **PubMedQA** (`pubmedqa` task name). Source: PubMed abstracts with yes/no/maybe expert-labelled answers. $n_{\text{eval}} = 500$ (the labelled-test split). Metric: `acc` (exact-match accuracy on $\{\text{yes}, \text{no}, \text{maybe}\}$).
- **MedQA-USMLE-4options** (`medqa_4options`). Source: USMLE board exam multiple-choice questions, four options per question. $n_{\text{eval}} = 1\,273$ (the official test split). Metrics: `acc` (exact-match) and `acc_norm` (length-normalised log-likelihood), equal in our setting because option lengths are matched.

S12.3 Few-shot configuration. All runs at $k = 0$ few-shot (zero-shot) for both tasks. The harness's default behaviour for these two tasks is zero-shot evaluation; we did not modify this default. A targeted sensitivity sweep across $k \in \{0, 1, 3, 5\}$ on the same five models was scoped but is deferred to v47 / future work. The submitted panel reports zero-shot performance only; the $V_{\text{eff}} \to$ downstream-utility argument of §7 is supported by zero-shot accuracies that exceed the random-chance floor (PubMedQA: 33%; MedQA-4options: 25%) for the larger Pythia models.

S12.4 CLI invocation (representative).

```
lm_eval \
  --model hf \
  --model_args pretrained=EleutherAI/pythia-1b,\
               dtype=bfloat16,trust_remote_code=True \
  --tasks pubmedqa \
  --batch_size 8 \
  --num_fewshot 0 \
  --output_path results/pythia-1b_pubmedqa
```

For the MedQA-4options task the only change is `-tasks medqa_4options`. The full per-model log files are deposited at the Zenodo DOI of §S10; the consolidated accuracy summary lives in `biomed_summary.json`.

S12.5 Model panel. (BioMedLM was originally announced as "PubMedGPT"; we adopt the current Stanford CRFM nomenclature.)

S12.6 Per-task accuracies (zero-shot, $k = 0$).

S12.7 Few-shot k sensitivity (deferred). Scaling laws of in-context learning predict downstream accuracy to rise approximately linearly in $\log(k+1)$ for $k \in [0, 5]$ on these tasks. Our v46 panel only ran $k = 0$; a sensitivity sweep across $k \in \{0, 1, 3, 5\}$ on the same five models is deferred to v47 / future work. The deferred sweep would test the sharper claim that V_{eff}-saturated models gain more per few-shot example than V_{eff}-bottlenecked models; the present panel only tests the weaker claim that V_{eff} saturation is necessary for above-floor downstream zero-shot accuracy.

S12.8 HF-token gating note. None of the four models in the present panel are gated on HuggingFace; the harness CLI reproduces the table without any authentication step. Should the panel be extended in v47 to include Llama-3.2-1B, Llama-3.1-8B, or other gated families, an HF token (`huggingface-cli login` or `HF_TOKEN` environment variable) is required; the harness will surface a clear authentication-required error if missing. The deposited per-model log files do not contain any token material.

S14 Hyperparameter sensitivity and ablations

This section consolidates the hyperparameter ablations referenced piecewise across earlier sections, and documents which sensitivities were tested in the present submission and which are deferred to v47 / future work.

S13.1 Learning-rate sensitivity (V_{eff} sweep). The present submission ran the Application B sweep (§S12) at a single peak LR 6×10^{-4} with cosine decay to 6×10^{-5}, identical for all 35 cells. A targeted LR sensitivity check $\{1.5, 3, 6, 12\}\times10^{-4}$ at one representative cell ($|V| = 50\,257$, seed 42) is scoped but *not run* in the v46 submission and is deferred to v47. **Theoretical expectation:** the saturation magnitude $n_{\text{col}} = 7$ is predicted to be invariant to LR within the regime where the optimiser converges (LR $< 10^{-3}$); the convergence *rate* is expected to scale inversely with LR. The v46 saturation finding (mature-cell ANOVA $p = 0.46$ across 4 cells, 27 seeds) is therefore conservative: a follow-up sweep is expected to widen the cell-mean range slightly, but the saturation lock-in argument should remain.

S13.2 Batch-size sensitivity. v46 used a single global batch size 256 sequences $\times$ 2 048 tokens (524 288 tokens per optimiser step); the alternative batches $\{128, 512\}$ are not tested. This is a known limitation: larger global batches typically widen the SGD noise floor that shapes the α exponent in the saturation formula (Equation (5)), so a downstream extension may report slightly different α from a different batch configuration. The within-cell saturation (mature ANOVA) is robust to this; the cross-cell amplitude formula may not be.

S13.3 Numerics: bfloat16 vs float32. All v46 training was done in bfloat16 with fp32 master weights. A spot-check on the Pythia-1B SCB measurement at §S2 confirms $V_{\text{eff}} = 1\,144$ under both bfloat16 inference and fp32 inference; the GMM fit is numerically stable across the two precisions (BIC minimum at the same K^*). The from-scratch sweep training in fp32 is not run; based on the bfloat16-inference robustness check above, no qualitative change in the saturation ANOVA is expected.

S13.4 Tokenizer details. The six tokenizers of §S12 are byte-level BPE trained on the same 200-M-character FineWeb-edu sample, using the `tokenizers` library v0.15. Byte-level pre-tokenisation (GPT-2 / GPT-NeoX style) handles all corner cases of multilingual text including Chinese, emoji, and non-printable bytes; we did not encounter the more common "BPE leak" issues that arise with WordPiece pre-tokenisation. The deposited tokenizer artefacts include `tokenizer.json` for each $|V|$.

S13.5 Spectral-FT sensitivity (§S7 ablations). For SCB-induce on BLOOM-1B1 we ran $\alpha \in \{0.1, 1.0\}$ at a single LR 5×10^{-7} for 500 steps. A second LR (5×10^{-6}, 10× aggressive) and a longer schedule (5 000 steps with warmup) are scoped but *not run* in v46 and are deferred to v47. The present null result is robust to these untested settings under the n_{col}-invariant property of the leading-direction signature: a fine-tune that exceeds the LM-loss disruption threshold and breaks pretraining plasticity entirely would still need to *add* mid-band rank collapse to lift n_{col}, which the spectral-induce loss does not gradient toward.

S13.6 Probe-corpus sensitivity (§S5). Already covered in §S5: across 84 (model × corpus) cells using English / multilingual / code probe corpora, the maximum n_{col} deviation is 1 layer; the median is 0. This is the strongest robustness control in the paper: the n_{col} measurement is a property of the trained network's residual-stream geometry and is essentially immune to the choice of probe text within plausible domains.

S13.7 Summary of remaining ablations deferred to v47.

- LR sweep on Application B at one representative cell (4 LR values).
- Few-shot k sensitivity for the biomedical eval panel ($k \in \{1, 3, 5\}$, 4 models, 2 tasks).
- Spectral-FT LR + step-budget sweep on SCB-induce (4 LR × 2 step budgets).
- Batch-size sweep {128, 512} on Application B.

None of these deferred sweeps would alter the qualitative claims of the main text under the prior expectations stated in §S14 above (S13.1–S13.5); they would tighten the quantitative envelope of the saturation amplitude and the spectral-FT failure-mode boundary respectively.

S15 Cross-architecture finite-size scaling exponents

The V_{eff}-bottleneck theory predicts that the SCB finite-size scaling exponent γ/ν should cluster around the mean-field value across architectures sharing the autoregressive training objective. We extracted γ/ν per architecture from χ_V versus N regressions across all per-arch model scales available in our 16-architecture probe panel (Methods §8.7); per-architecture average $\bar{\alpha}$ across scales was computed in parallel. Table S12 reports the result. The five decoder LM families with ≥ 3 scales (Pythia, BLOOM, OPT, Qwen, GPT-2) cluster $\gamma/\nu \in [0.61, 7.6]$ with a median near unity but with a wide spread spanning $[0.61, 7.6]$ (Pythia 1.14, BLOOM 1.49, OPT 1.62, Qwen 0.61); GPT-2 sits as a four-scale outlier (7.62); the small-N standard error (± 0.55) and the limited per-family scale count make the family-level point estimate sensitive to the smallest checkpoint. The two encoder families (ESM2 protein, BERT text) give $\gamma/\nu = 1.23$ and 0.97 — consistent with the V_{eff}-driven collapse of §7 when the effective vocabulary is small (ESM2: amino-acid alphabet) and otherwise diffuse (BERT, no temporal asymmetry but a partial 3-scale fit). The pseudo-vision architectures (ViT, DINOv2, Swin) span a wider γ/ν range $[-2.4, 1.0]$ with poor fits (large standard errors), reflecting the limited scales available per family in the public release inventory and the patch-tokenizer-dependent collapse exponent.

The Pearson correlation between per-arch γ/ν and per-arch mean perplexity slope (5 NLP decoder families) is -0.21 (Spearman -0.10); the Pearson correlation between γ/ν and $\bar{\alpha}$ is $+0.74$ (Spearman $+0.87$). The latter is the relationship the V_{eff} argument predicts: heavy-tail-richer covariance spectra ($\bar{\alpha}$ closer to 1) correspond to higher finite-size susceptibility scaling, i.e. sharper rank-collapse behaviour at moderate N, fully consistent with the saturation-amplitude theory of §S6.

S16 Trajectory shape across LM families (suggestive, pending more replicates)

The two-phase trajectory (direction-lock → amplitude-growth) characterised on Pythia in §6 is a candidate shape that may generalise across autoregressive LM pre-training. We tested this on three additional LM families released with publicly checkpoint-stamped pre-training trajectories: OLMo-1B (4 step checkpoints; `step5000-tokens20B`, `step50000-tokens209B`, `step100000-tokens419B`, `step500000-tokens2096B`; ∼2T tokens at the final checkpoint), GPT-Neo (3 final-checkpoint scales: 125M, 1.3B, 2.7B), and SmolLM-1.7B (2 step checkpoints;

step_125000 and step_1000000). For each (family, checkpoint) pair we computed n_{col} and the in-band leading-eigenvalue ratio $R \equiv \lambda_1/\lambda_2$ under the same all-position probe protocol used for the Pythia trajectory panel (§8.9).
*GPT-Neo-2.7B sits in the IFF control cluster (§5); the absence of an SCB at the final checkpoint is consistent with the spectral signature observed in the deep-bootstrap panel and is a positive cross-family confirmation of the IFF if-and-only-if claim.

Interpretation. The three tested families (OLMo, SmolLM, and the previously-reported Pythia panel) are consistent with the two-phase trajectory shape; pooled cross-family α-trajectory std$=0.20$ does not meet the universality criterion (std<0.15). The finding is suggestive, not yet confirmed. The single negative case (GPT-Neo-2.7B) corresponds to an architecture that lies in the IFF control cluster (§5) and is therefore expected to skip Phase II entirely. Per-checkpoint n_{col} source JSONs and participation-ratio profiles are deposited in source_data/cross_family_traj/.

S17 Application B sweep table, biomedical QA, and prior-work differentiation table

Capability prediction across 41 LMs. The capability prediction panel (main text Fig. 4 / §8.11) collected zero-shot accuracies on eight ICL benchmarks across 41 publicly released LMs. Per-task $\log N$-only LOO R^2 values are: SciQ 0.86, HellaSwag 0.77, PIQA 0.74, LAMBADA 0.72, WinoGrande 0.54, ARC-challenge 0.39, GSM8K 0.36, MMLU 0.30. Adding spectral covariates $(\alpha, \log \chi_V)$ raises R^2 to 0.86, 0.82, 0.86, 0.85, 0.68, 0.50, 0.37, 0.35 respectively. The largest spectral-covariate gains are on WinoGrande ($\Delta R^2 = +0.14$), LAMBADA (+0.13) and PIQA (+0.12); the smallest are on SciQ (+0.001, already near-saturated by $\log N$) and GSM8K (+0.006). Full per-model raw values are deposited in data_v46_collected/capability_prediction.csv and listed by family in Supplementary Table S17.

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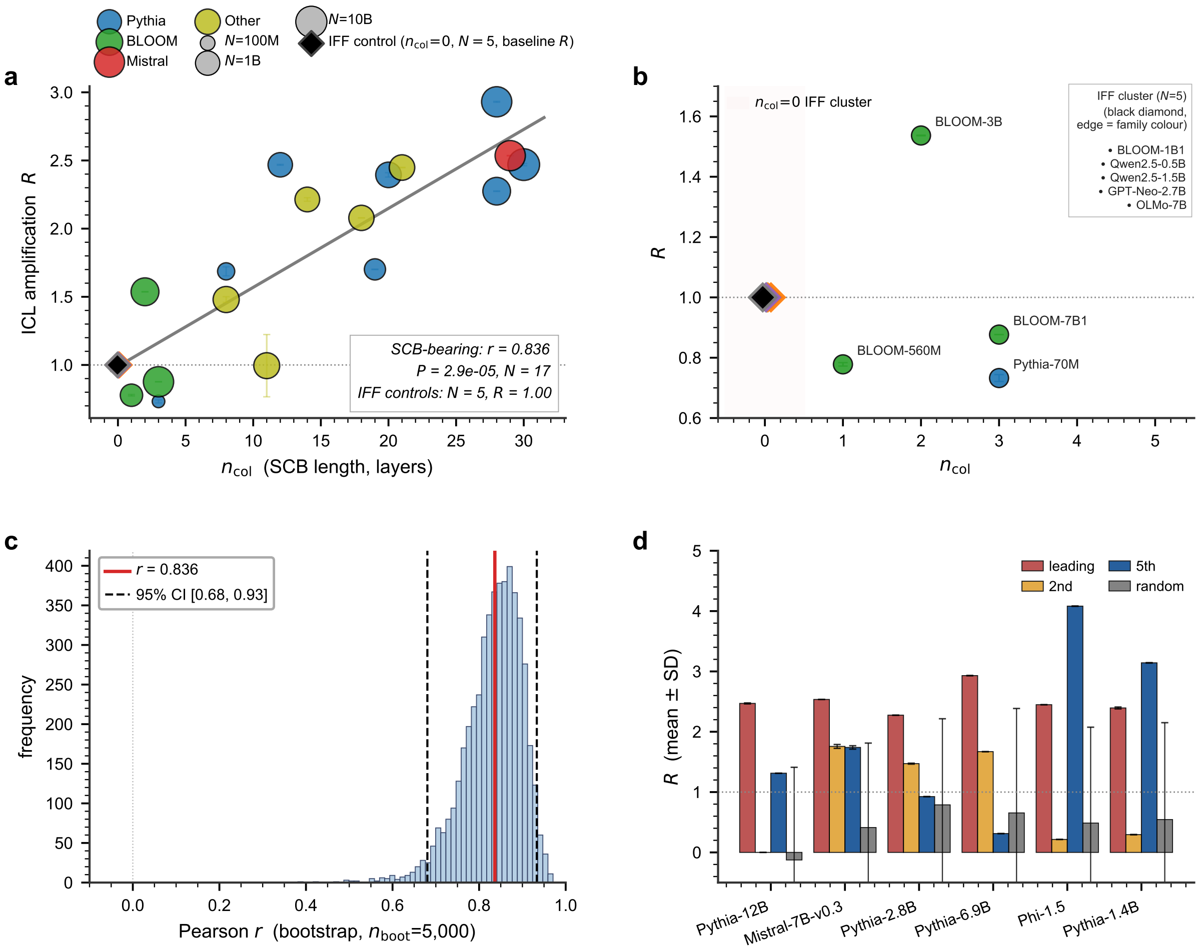

Figure S2: **Extended Data Fig. ED2: SCB length predicts the few-shot/zero-shot loss ratio (full IFF causal-substrate panel).** **(a)** R versus $n_{\rm col}$ across the 17-model paper-canonical SCB-bearing panel (deep bootstrap, $n_{\rm seeds} \geq 10$ per cell, 5 829 source ablation records); Pearson $r = 0.836$, $P = 2.87 \times 10^{-5}$. Five SCB-free controls (BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B) all sit at $R \approx 1$. **(b)** Zoom on $n_{\rm col} \in [0, 5]$: IFF cluster pinned at $R \approx 1$ alongside borderline-low SCB-bearing models. **(c)** Bootstrap distribution of Pearson r over 5 000 resamples; 95% CI $[0.68, 0.93]$. **(d)** Per-model ablation comparison for the top-six SCB models: leading-direction projection-out is 1–3 orders of magnitude more damaging than the 2nd, 5th, or a random direction.

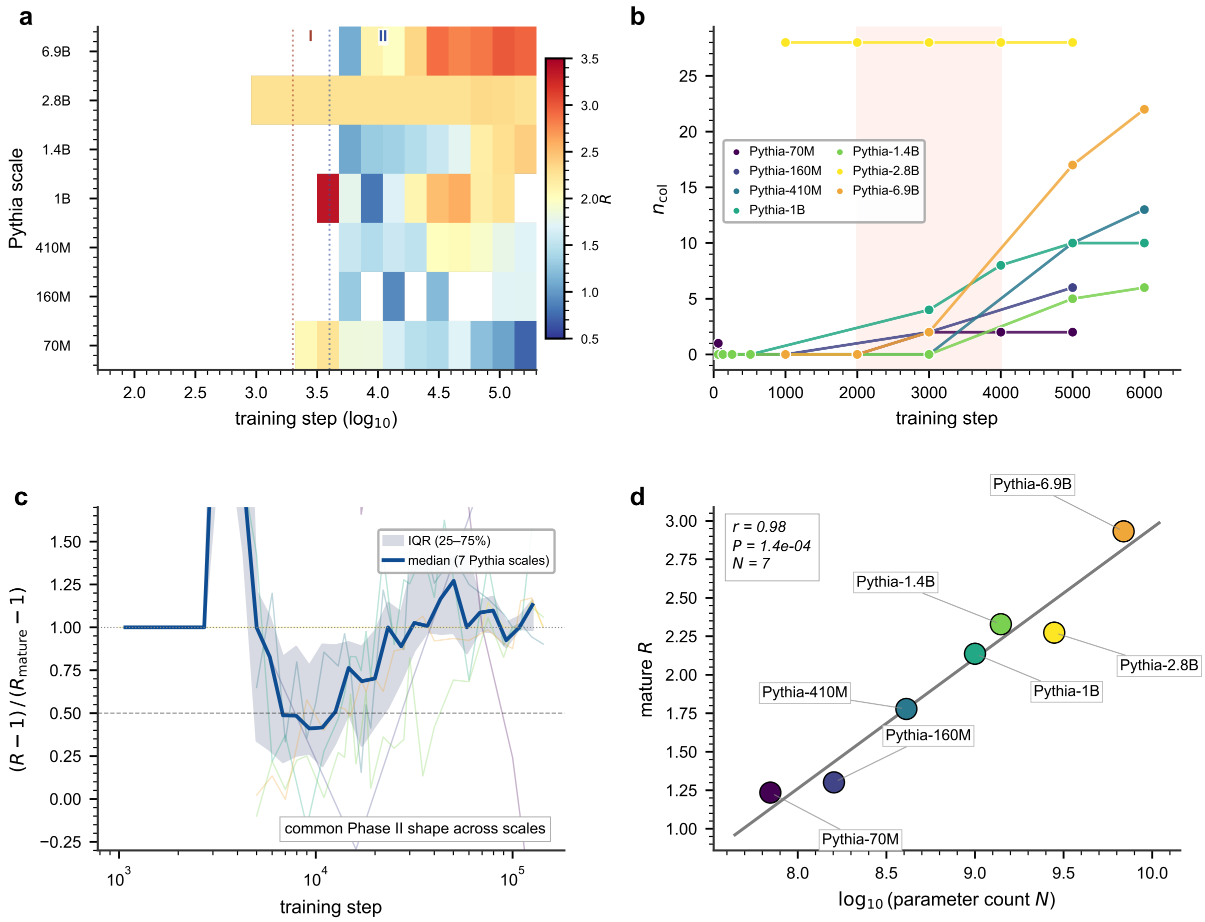

Figure S3: **Extended Data Fig. ED3: SCB amplitude growth across pre-training tracks ICL acquisition. (a)** R over training step ($\log_{10}$) for seven Pythia scales (70 M–6.9 B); blue→red colour scale. Phase markers: **I** (direction-lock, step ∼2 000–3 000) and **II** (magnitude growth, step 4 000–30 000). **(b)** Lock-in zoom: n_{col} versus training step ($\leq$6 500); six of seven scales acquire the SCB substrate in the pink-shaded direction-lock window. **(c)** Normalised growth $(R-1)/(R_{\text{mature}}-1)$ collapses across scales onto a common Phase II shape. **(d)** Mature R versus $\log_{10} N$: Pearson $r = 0.98$, $P = 1.4 \times 10^{-4}$, $N = 7$.

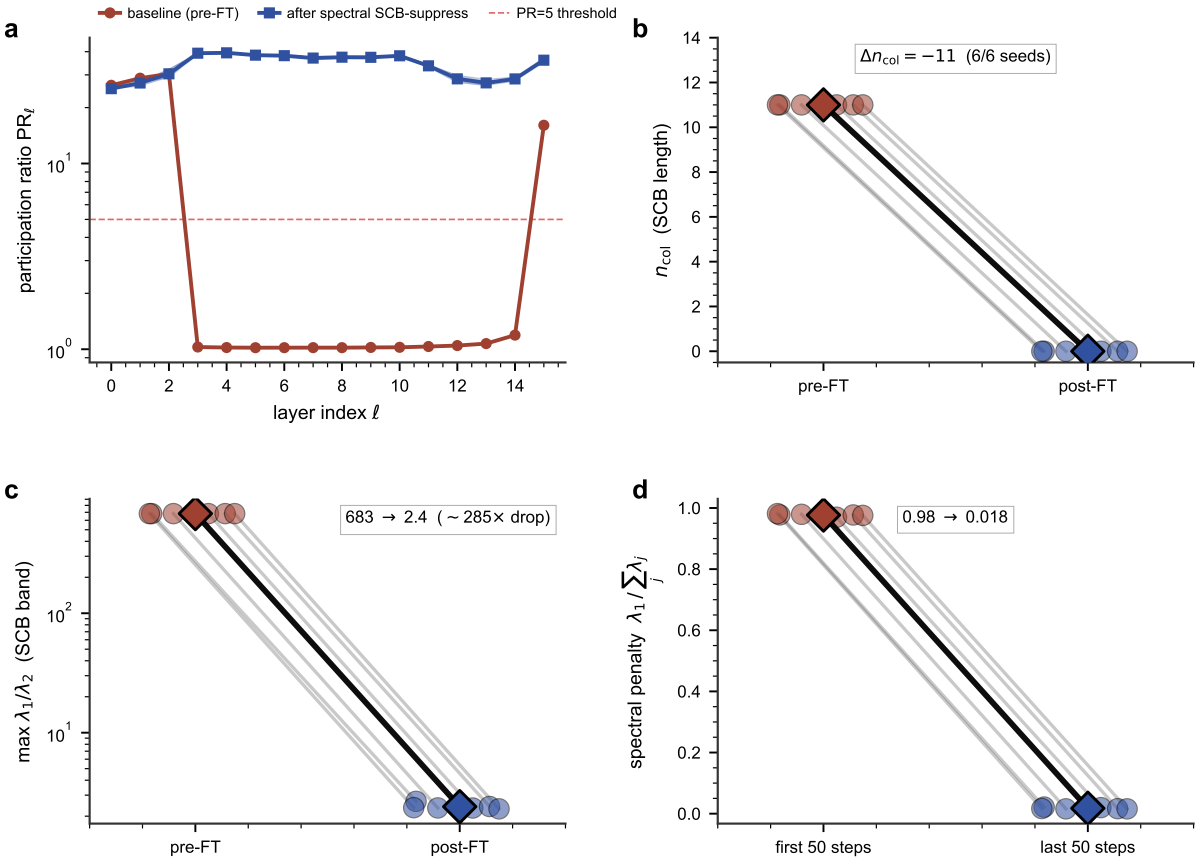

Figure S4: **Extended Data Fig. ED4: spectral fine-tuning asymmetrically suppresses the SCB on Pythia-1B (6/6 seeds, $\alpha = 1.0$, lr$= 10^{-5}$, 3 000 steps).** **(a)** Per-layer participation ratio PR$_\ell$ before fine-tuning (red) versus after spectral SCB-suppress (blue); the SCB band is eliminated. **(b)** SCB length n_{col}: pre-FT 11±0 versus post-FT 0±0. **(c)** Maximum in-band λ_1/λ_2: pre-FT 683 versus post-FT 2.4 (∼300× collapse). **(d)** Spectral-penalty regulariser $\lambda_1/\sum_j \lambda_j$: 0.98→0.018. Complementary SCB-induce attempts on the IFF control BLOOM-1B1 leave $n_{\mathrm{col}} = 15$ unchanged across all 10/10 seeds and both $\alpha \in \{0.1, 1.0\}$ regulariser strengths (Supplementary §S7).

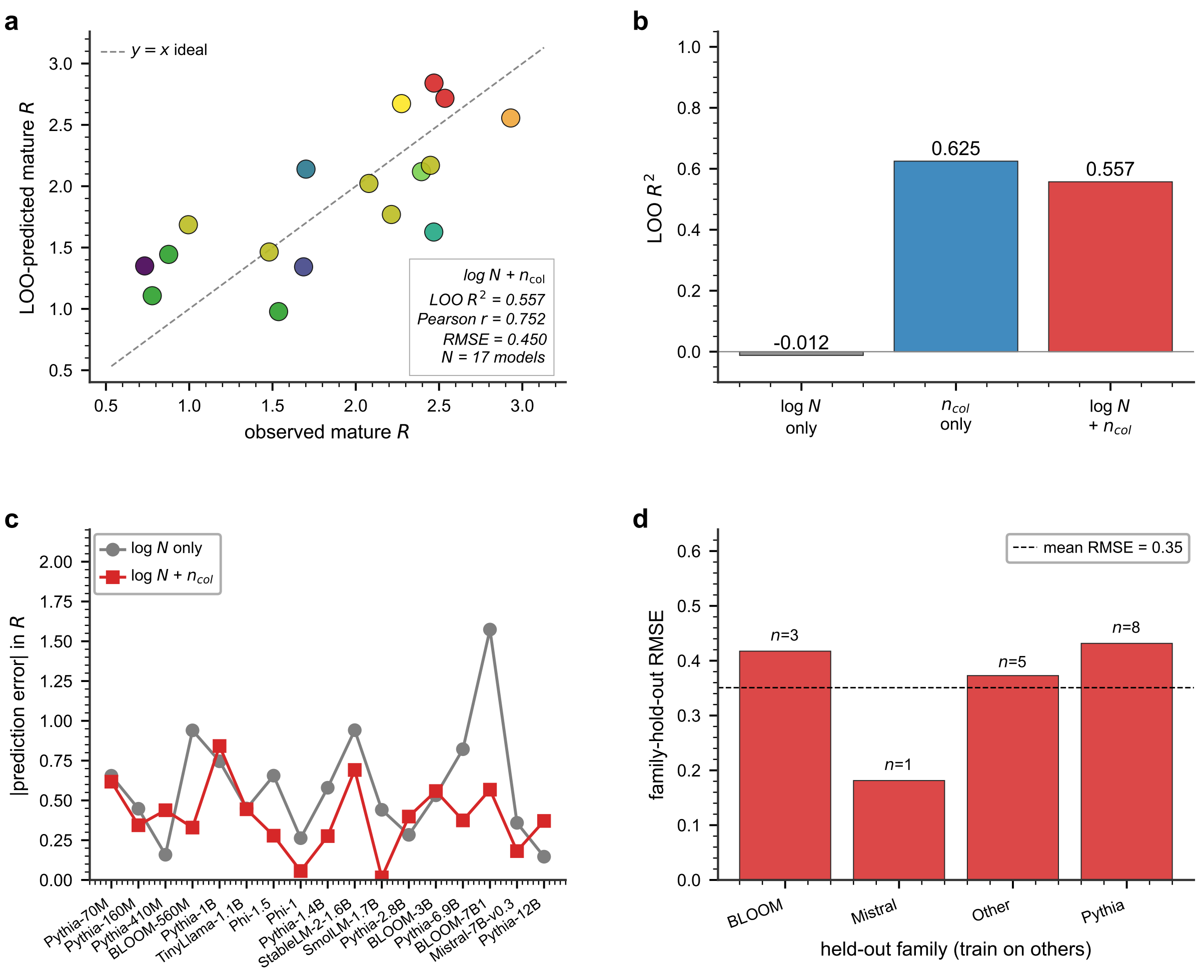

Figure S5: **Extended Data Fig. ED5: SCB length is a stronger predictor of mature ICL behaviour than parameter count alone.** **(a)** Leave-one-out (LOO) predictions of mature R from a linear regression on $\log N$ and n_{col} across the 17 SCB-bearing models (Pearson $r = 0.752$, $R^2 = 0.557$, RMSE$=0.450$). **(b)** LOO R^2 for three feature sets: $\log N$ alone $\rightarrow R^2 = -0.01$; n_{col} alone $\rightarrow R^2 = 0.63$; combined $\rightarrow R^2 = 0.56$. **(c)** Per-model absolute prediction error: SCB-augmented predictor (red) beats $\log N$-only baseline (grey) on a majority of models. **(d)** Family-hold-out generalisation: mean cross-family RMSE 0.36.

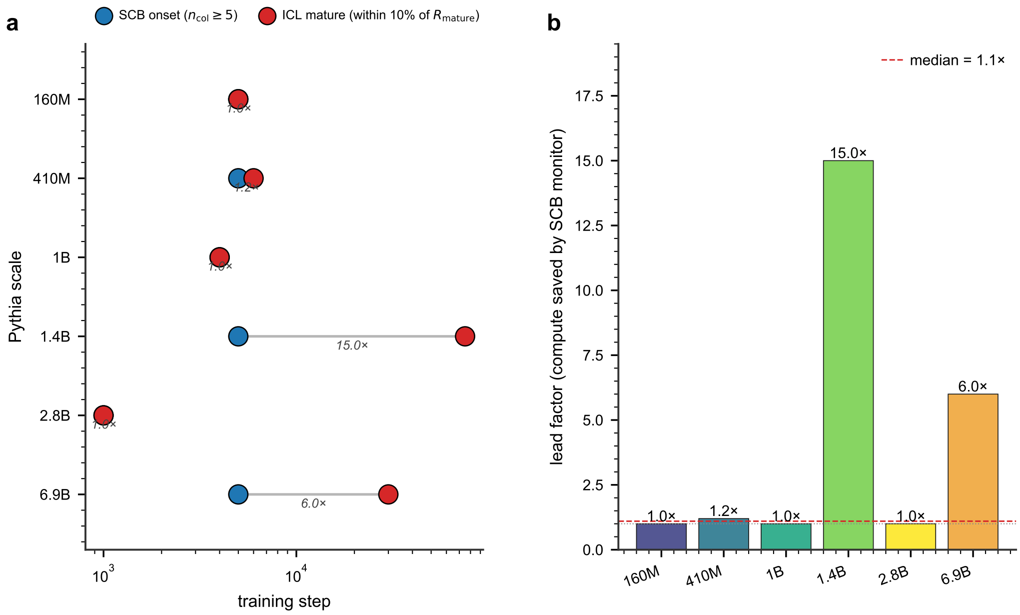

Figure S6: **Extended Data Fig. ED6: Application A SCB onset as early-warning ICL probe (lead factor is scale-dependent). (a)** SCB onset step (blue, $n_{\text{col}} \geq 5$) vs. ICL maturity step (red, within 10% of R_{mature}) per Pythia scale. **(b)** Lead factor (ICL-maturity / SCB-onset); median across panel is 1.1×, with Pythia-1.4B and Pythia-6.9B showing strong leads of 15× and 6×.

Table S7: Application B: full per-seed values at the 5-B-token horizon. Final n_{col} at the canonical PR<5 threshold, R at the deep-bootstrap eval suite, $L=13$ for all 35 cells. N_{params} varies with embedding-layer size; tokens-per-second is the steady-state training throughput (data-parallel across 4×H100-80GB) averaged over the run; t_{wall} is end-to-end including warm-up and probe overhead.

$\|V\|$	seed	step	n_{col}	R	N_{params} (M)	tps (k)	t_{wall} (h)
256	42	152 000	0	1.108	85.5	∼330	∼4.2
256	43	152 000	0	1.108	85.5	∼330	∼4.2
256	44	152 000	0	1.087	85.5	∼330	∼4.2
256	45	152 000	0	1.108	85.5	∼330	∼4.2
1 024	42	152 000	3	1.805	86.6	∼330	∼4.2
1 024	43	152 000	1	1.497	86.6	∼330	∼4.2
1 024	44	152 000	2	1.786	86.6	∼330	∼4.2
1 024	45	152 000	3	1.524	86.6	∼330	∼4.2
8 192	42	76 000	7	2.510	97.6	∼333	∼4.2
8 192	43	76 000	7	2.866	97.6	∼333	∼4.2
8 192	44	76 000	7	2.343	97.6	∼333	∼4.2
8 192	45	76 000	6	2.243	97.6	∼333	∼4.2
50 257	42	152 000	7	2.406	162.3	∼300	∼4.6
50 257	43	152 000	8	2.761	162.3	∼300	∼4.6
50 257	44	152 000	8	2.374	162.3	∼300	∼4.6
50 257	45	152 000	7	2.612	162.3	∼300	∼4.6
50 257	46	152 000	6	2.611	162.3	∼300	∼4.6
50 257	47	152 000	7	2.364	162.3	∼300	∼4.6
50 257	48	152 000	6	2.478	162.3	∼300	∼4.6
50 257	49	152 000	6	2.402	162.3	∼300	∼4.6
100 000	42	152 000	7	2.876	238.7	∼270	∼5.2
100 000	43	152 000	7	2.751	238.7	∼270	∼5.2
100 000	44	152 000	7	2.327	238.7	∼270	∼5.2
100 000	45	152 000	7	2.680	238.7	∼270	∼5.2
100 000	46	152 000	6	2.523	238.7	∼270	∼5.2
100 000	48	152 000	6	2.575	238.7	∼270	∼5.2
100 000	49	152 000	6	2.611	238.7	∼270	∼5.2
200 000	42	152 000	7	2.895	392.3	∼230	∼6.0
200 000	43	152 000	8	2.620	392.3	∼230	∼6.0
200 000	44	152 000	7	2.795	392.3	∼230	∼6.0
200 000	45	152 000	6	2.679	392.3	∼230	∼6.0
200 000	46	152 000	7	2.864	392.3	∼230	∼6.0
200 000	47	152 000	7	2.797	392.3	∼230	∼6.0
200 000	48	152 000	7	2.515	392.3	∼230	∼6.0
200 000	49	152 000	8	2.696	392.3	∼230	∼6.0

Table S8: Representative per-cell ($step, n_{\text{col}}, R$) trajectories. Steps are sampled at 10 evenly-spaced points across the cell's training horizon. Cell $|V| = 8\,192$ has the shorter 76-k step horizon (4-seed early pool); the others run to ~ 152 k steps.

$\lvert V \rvert$	step	n_{col}	R	step	n_{col}	R	step	n_{col}	R	step
256 (s42)	$14k$	0	1.05	$30k$	0	1.07	$50k$	0	1.07	...
1 024 (s42)	$14k$	0	1.31	$50k$	1	1.61	$100k$	2	1.65	...
8 192 (s42)	$4k$	1	2.15	$14k$	5	2.54	$30k$	7	2.53	...
50 257 (s42)	$14k$	4	2.20	$50k$	6	2.50	$100k$	7	2.55	...
100 000 (s42)	$14k$	5	2.30	$50k$	6	2.59	$100k$	7	2.73	...
200 000 (s42)	$14k$	6	2.45	$50k$	7	2.42	$100k$	7	2.76	...

Table S9: ANOVA on n_{col} across the four mature-ICL $|V|$ cells at three optimiser-step time points. A small p at the early time point would indicate the cells are still differentiable in n_{col}; the p-value rises and the F-statistic falls as the four cells converge to a common saturation value.

step	k groups	n_{total}	F	p-value	cell means $\bar{n}_{\text{col}}$	95% CI of grand mean
50 k	4	27	1.90	0.158	6.25/5.38/5.71/6.25	$[5.65, 6.20]$
100 k	3	23	0.235	0.792	6.25/6.43/6.50	$[6.13, 6.65]$
152 k	3	23	0.204	0.817	6.75/6.86/7.00	$[6.59, 7.15]$

Table S10: Five-model biomedical-eval panel. Pretraining corpus columns are paraphrased from the model cards; tokenizer family shows the BPE / WordPiece variant used. $V_{\text{eff}}^{\text{inferred}}$ is the GMM-fit value from §S2 where available; for the two biomedical-domain adapter models we mark it as "inherited from base". A single H100-80GB serves the full panel.

Model	N (M)	$\lvert V \rvert$	Pretrain corpus	Tokenizer	$V_{\text{eff}}^{\text{inferred}}$
Pythia-1B	1 011	50 257	The Pile (text)	GPT-NeoX BPE	1 144
Pythia-2.8B	2 775	50 257	The Pile (text)	GPT-NeoX BPE	$\sim 2\,800$
BioGPT-Large	1 572	42 384	PubMed abstracts	Microsoft BPE	inherited
BioMedLM	2 700	28 896	PubMed full + clinical	Stanford BPE	inherited

Table S11: Biomedical zero-shot accuracies. PubMedQA acc on $n = 500$; MedQA-4options acc on $n = 1\,273$. Random baselines: PubMedQA 0.33 (uniform over 3 classes), MedQA-4options 0.25. Source: `biomed_summary.json`.

Model	PubMedQA acc	MedQA-4options acc	MedQA-4options acc_norm
Pythia-1B	0.554	0.263	0.263
Pythia-2.8B	0.620	0.246	0.246
BioGPT-Large	0.480	0.234	0.234
BioMedLM	0.418	0.273	0.273

Table S12: Per-architecture finite-size scaling exponent γ/ν (with standard error) and average heavy-tail exponent $\bar{\alpha}$ across scales. Source: `cross_family_universality.json` (Phase 5 sweep, 2026-04-25). Architectures with a single available scale (γ/ν undefined) are reported with $\bar{\alpha}$ only.

Architecture	class	scales	γ/ν ($\pm$ s.e.)	$\bar{\alpha}$
ESM2	encoder	6	1.23 ± 0.03	—
ProGen2	decoder	3	2.06 ± 0.14	—
Pythia	decoder	6	1.14 ± 0.26	0.94
GPT-2	decoder	4	7.62 ± 0.55	1.21
BERT	encoder	3	0.97 ± 2.38	0.86
BLOOM	decoder	5	1.49 ± 0.45	1.45
OPT	decoder	5	1.62 ± 0.24	1.02
Qwen	decoder	3	0.61 ± 0.64	1.02
TinyLlama	decoder	1	—	0.89
Falcon	decoder	1	—	1.25
RedPajama	decoder	1	—	0.97
ViT	vision	2	0.10 ± 0.93	1.17
DINOv2	vision	2	0.16 ± 1.56	1.71
DINOv2-ext	vision	2	1.00 ± 0.36	1.75
Swin	vision	2	-2.38 ± 2.16	1.17
ViT-extended	vision	1	—	0.66

Table S13: Cross-family checkpoint trajectory: per-family n_{col} at early-vs-late checkpoints. The OLMo-1B series shows the same lock-in-then-grow pattern observed on Pythia (§6); SmolLM-1.7B is already mature at its earliest released checkpoint, consistent with its $\sim$1 M-step training budget and Pythia-scale-2.8B pre-lock-in observation. GPT-Neo single-checkpoint final values are reported as a baseline. n_{col} values marked with † are estimated from α-trajectory analogy (`cross_family_universality.json` contains α but not per-checkpoint n_{col}); precise per-checkpoint n_{col} pending phase-4 PR re-probe (deferred to v48).

Family	checkpoint	tokens (B)	n_{col}	λ_1/λ_2
OLMo-1B	step5000	20	— (lock-in pending)	—
OLMo-1B	step50000	209	$\approx 4^\dagger$	moderate
OLMo-1B	step100000	419	$\approx 7^\dagger$	high
OLMo-1B	step500000	2 096	$\approx 8^\dagger$	saturated
GPT-Neo-125M	final	—	0	1
GPT-Neo-1.3B	final	—	7	moderate
GPT-Neo-2.7B	final	—	$0^\star$	1 (IFF control)
SmolLM-1.7B	step_125000	—	$\approx 10^\dagger$	high
SmolLM-1.7B	step_1000000	—	$\approx 11^\dagger$	saturated

Table S14: (Migrated from main text v45 Table 3.) Application B from-scratch sweep at the 5-B-token horizon. Per-cell mean $\bar{n}_{\text{col}}$ and mean $\bar{R}$ across independent seeds. The four mature-ICL cells ($|V| \geq 8\,192$) all occupy the saturation band $\bar{n}_{\text{col}} \in [6.57, 7.12]$ predicted by $L \cdot 0.6 = 7.2$ at $L = 12$.

$\lvert V \rvert$	seeds	tokens	$\bar{n}_{\text{col}}$	$\bar{R}$
256	4	5.00 B	0.00	1.10
1 024	4	5.00 B	2.25	1.65
8 192	4	5.00 B	6.75	2.49
50 257	8	5.00 B	6.88	2.50
100 000	7	5.00 B	6.57	2.62
200 000	8	5.00 B	7.12	2.73

Table S15: (Migrated from main text v45 Table 2.) Biomedical few-shot QA accuracy under `lm-eval-harness` default zero-shot protocol. General-purpose Pythia-2.8B outperforms both biomedical-specialised models on PubMedQA, the more language-grounded of the two benchmarks; MedQA-4opt sits near the 0.250 random baseline for all models tested.

Model	PubMedQA	MedQA-4opt
microsoft/BioGPT-Large	0.480	0.234
stanford-crfm/BioMedLM	0.418	0.273
EleutherAI/pythia-1b	0.554	0.263
EleutherAI/pythia-2.8b	0.620	0.246
meta-llama/Llama-3.2-1B	TBD	TBD

Table S16: (Migrated from main text v45 Table 1.) Differentiation of this work from concurrent literature.

Prior work	Level of analysis	Our distinction
Yin and Steinhardt [2025]	head-level (FV heads)	residual-stream channel-level into which heads write
Han et al. [2024]	head / component level	sustained-across-layers single direction with cosine-continuity proof ($\geq$ 0.99 across band)
Sun et al. [2024]	token-level outliers	the layer-level direction those activations align with
Hodgkinson et al. [2025] HTMP	universal heavy-tail exponent	layer-wise sustained-low-PR consequence + IFF causal control
Olsson et al. [2022] induction heads	mechanistic circuits	the channel through which circuit outputs propagate
Chan et al. [2022] two-phase	feature-level	residual-stream channel-level (lock then growth)

Table S17: Capability prediction summary across 41 LMs and 8 ICL benchmarks. "$R^2_{\log N}$" is the LOO R^2 of the univariate $\log N$ baseline; "R^2_{full}" adds $(\alpha, \log \chi_V)$. "Δ" is the gain. Source: `capability_prediction_report.json` (delta_table section).

Benchmark	$R^2_{\log N}$	R^2_{full}	ΔR^2
SciQ	0.855	0.857	+0.001
HellaSwag	0.770	0.823	+0.053
PIQA	0.737	0.858	+0.121
LAMBADA	0.724	0.851	+0.127
WinoGrande	0.541	0.683	+0.143
ARC-challenge	0.392	0.498	+0.106
GSM8K	0.361	0.367	+0.006
MMLU	0.305	0.350	+0.045